The cost of precision-fermented whey and egg proteins

A Humbird-style techno-economic analysis of β-lactoglobulin and ovalbumin, from stoichiometry to the physical limit, with a lactoferrin contrast case

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Status. Not yet peer-reviewed. The model code, the evidence extraction and this text were produced with an AI assistant (Claude, Anthropic) under the author's direction. Version 1 was audited by eight independent AI reviewers, each owning one area and required to support every finding with a computation, quote or code line; every finding, decision and effect is logged in `docs/AUDIT.md`, and this version implements them. No human expert has reviewed the analysis yet. Every number is computed by the model and read from `results/analyses.json` at compile time; every input traces to a source and an evidence grade in `data/` (Appendix A). Inputs graded E are guesses; they are listed in Section 6.

Summary

Precision fermentation (PF) uses an engineered microbe to secrete a single animal protein. We estimate what food-grade β-lactoglobulin (BLG, the main whey protein) and ovalbumin (OVA, the main egg-white protein) cost to make, and what performance would bring them to the price of the proteins they replace. The method is Humbird's (2021): element- and electron-balanced black-box stoichiometry, an oxygen-transfer-limited fed-batch fermenter, and a capital estimate tied to the market price of baker's yeast. We add a secreted product, a downstream train and a cation-exchange step, and check the model against three published designs.

The base case is a secreting **Komagataella phaffii** strain at 30 g/L after a 72 h production phase, the design point of Intelligen's (2025) OVA process, in 200 m³ fermenters making 10 kt/yr of protein in the US Midwest (2025 dollars). With a capital charge of 14.6% of investment per year its minimum selling price is **$29.4/kg BLG** and **$29.5/kg OVA**. With the lighter capital convention of most published PF studies it is $16.1 and $16.1. At the private-sector average titer reported to GFI (42 g/L) it is $21.9/kg. At the best public titers it is $64/kg BLG (13.9 g/L, with an ethanol-fed production phase that the model prices as glucose) and $390/kg OVA (2 g/L). Whey protein traded at $9–$18 per kg protein in 2015–2025 (export unit values in 2025 dollars) and dried egg white at $10–$21.

Four results stand out:

1. **A feasibility ceiling on titer, set by the vessel's volume.** A higher titer needs more sugar feed, and with a 50% glucose feed and a batch that must start at least 30% full, a 200 m³ fermenter reaches about 74 g/L after a 72 h production phase. Both numbers are design choices: with a 20% start or a 65% feed the ceiling rises to about 125 g/L, where the secretion rate or O_2 transfer binds instead. Inside the ceiling, $20/kg BLG needs 40–47 g/L in 48–72 h.
2. **A physical-limit floor** of $6.1/kg BLG and $6.2/kg OVA ($3.6 and $3.6 with the lighter capital convention) at 100 kt/yr, with every biological parameter at its best measured or theoretical value at once.
3. **Capital conventions are the largest structural uncertainty.** On the same processes, published costs are 2.4–2.5× lower than ours, because of lighter capital and fixed-cost conventions, not different biology.
4. **A high-value contrast.** The same model puts recombinant lactoferrin at $216/kg ($134 with lighter capital) at the design titer of 10 g/L, against $200–$1,500/kg reported for dairy lactoferrin: below most reported prices. For high-value proteins PF can already compete on cost; the commodity proteins are the hard case, and whey, a by-product of cheese, is the hardest price to beat.

Scenario	Titer (g/L)	BLG, price-anchored	BLG, design-report	OVA, price-anchored	OVA, design-report
Best public academic strain	13.9 / 2	$64	$32	$390	$167
Base case (72 h production)	30	$29.4	$16.1	$29.5	$16.1
Industry benchmark (GFI private average)	42	$21.9	$12.1	$22.0	$12.1
Physical limit (100 kt/yr)	119 / 118	$6.1	$3.6	$6.2	$3.6
Incumbent, per kg protein (2015–2025, 2025$)			whey $9–$18		egg white $10–$21

Table 1: Minimum selling price ($/kg target protein, 2025) with a capital charge of 14.6% of total capital investment per year. The two capital conventions are defined in Section 2.5. 10 kt/yr and US Midwest unless noted. Academic BLG: Yao et al. (2026), whose production phase runs on ethanol (priced as glucose); on glucose, Li et al. (2026) reach 5.14 g/L ($164/kg).

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1 Question and approach

PF companies now sell BLG (Perfect Day, Remilk, Vivici) and OVA (EVERY), and others are entering (Onego Bio). Whether these proteins can displace animal proteins at scale or stay a premium niche depends on what they cost to make. Published estimates disagree by a factor of several, and company claims are rarely checked against the physics of aerobic fermentation. This study asks:

> What does it cost to make food-grade BLG and OVA by precision fermentation with today's public strain performance, and at the physical limits of aerobic fermentation? What combination of titer, rate, yield, downstream design and scale would bring them to price parity with whey and egg-white proteins?

#	Sub-question	Answered in
Q1	What do BLG and OVA cost at public (academic) and claimed (industry) performance?	Section 4.2, Figure 2
Q2	What drives the cost, and how good is the evidence for each driver?	Section 4.3, Figure 3, Figure 4, Figure 5
Q3	Which combinations of performance reach $10 and $20 per kg protein?	Section 4.4, Figure 6
Q4	What is the cost at the physical limits?	Section 4.5, Figure 7
Q5	Are published company claims consistent with the fermenter's limits?	Section 4.6, Table 8
Q6	How much do siting (US vs EU) and scale (1–100 kt/yr) matter?	Section 4.7, Figure 8
Q7	*Added after v1:* what do these costs mean per litre of milk and per egg?	Section 4.8, Figure 9
Q8	*Added after v1:* how does a high-value protein (lactoferrin) compare?	Section 4.9, Figure 10

Table 2: Q1–Q6 were fixed before any model output existed (`docs/SCOPE.md`); Q7–Q8 were added at the author's request after the first draft.

Why whey and egg white are the benchmarks. BLG is the main protein of whey, and the PF BLG on the market is sold as a whey-protein ingredient for bars, drinks and dairy analogues (FDA GRN 1056 specifies a concentrate of at least 70% protein). A buyer's alternative is whey protein concentrate or isolate, so their price is the one PF BLG must meet; the same logic sets dried egg white against OVA. This is the hardest possible comparison. Whey is a by-product of cheese, sold for whatever clears the market once cheese has paid for the milk, and it is among the cheapest animal proteins. Two consequences follow. Price parity is not the only commercial route: pure, animal-free or functional proteins can sell at a premium, as the lactoferrin case shows (Section 4.9). And displacing whey changes milk output far less than one-for-one, because whey carries only a small part of the milk's value (Section 4.8).

Why Humbird's method. Humbird's (2021) cultured-meat analysis builds costs from conservation laws: element and electron balances, oxygen transfer, heat removal. Its capital estimate reproduces the market price of one fermented commodity, baker's yeast at about $1.90/kg. We port his baker's-yeast workbook exactly (Section 3) and extend it with a secreted protein, a downstream train, and the specifics of **K. phaffii** and **Trichoderma reesei**.

Rules we followed (written protocol, `PLAN.md`):

- Four scenarios are never merged: academic, industry, required and physical limit.
- Every input carries an evidence grade, A to E.
- The model structure is frozen before the calibration anchors are run; misses are recorded, not tuned away.
- Priors about the results were written down before the model existed; they are scored in Section 7.
- Model-to-model agreement is a consistency check, not validation.
- After the first draft, eight independent reviewers audited it; their findings and every resulting change are logged in `docs/AUDIT.md` (R1–R8, P-1 to P-13).

2 Methods

2.1 Plant, products and functional unit

The functional unit is 1 kg of target protein in a spray-dried powder that meets the product specification. The base specification is a **concentrate**, following Remilk's GRAS notice for BLG (FDA GRN 1056): at least 70 wt% protein, at least 80% of it BLG, at most 7% moisture. OVA uses the same specification by analogy. The **isolate** variant (99% target protein, dry basis) adds an ammonium-sulfate precipitation, following Intelligen's (2025) OVA design. The lactoferrin contrast case follows the recombinant bovine lactoferrin notices (FDA GRN 1219 and 1284): at least 90% protein, at least 95% of it lactoferrin.

The plant is greenfield, single-product and nth-of-a-kind, built to food (not pharmaceutical) good manufacturing practice. The base is 10 kt/yr of protein in the US Midwest, swept from 1 to 100 kt/yr, with a north-west Europe (Netherlands) variant. Money is 2025 US dollars. Excluded costs: R&D, strain licences, regulatory approval, sales and distribution, land, first-of-a-kind premiums and carbon pricing. Figure 1 shows the base-case plant.

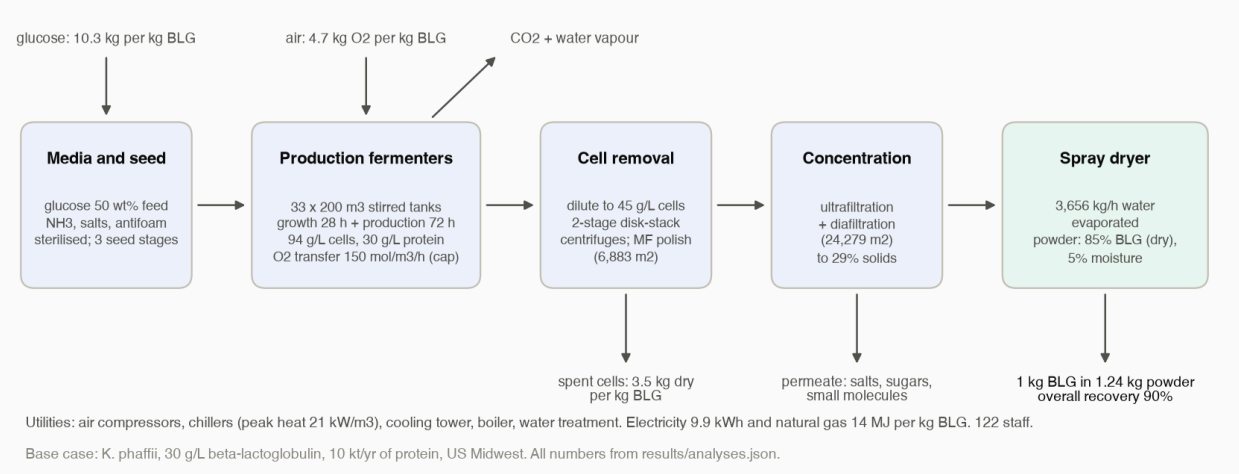


Figure 1: Base-case process: **K. phaffii** secreting BLG at 30 g/L; flows per kg of BLG.

2.2 Stoichiometry and energetics

Each metabolic process is one element-balanced "black-box" reaction on C, H, O and N: glucose + NH_3 + O_2 ⟶ biomass (or protein) + CO_2 + H_2O. Electrons are tracked through the degree of reduction ɣ per C-mol (glucose 4, **K. phaffii** biomass 4.06, BLG 4.24, OVA 4.18). This closes the O_2 and CO_2 balances once the carbon split is fixed.

Growth. Biomass yield comes from the Gibbs-energy dissipation correlation of Heijnen and van Dijken (1992), as in Humbird's workbook. The energy dissipated per C-mol of biomass formed is

$$D = 200 + 18(6 - c)^{1.8} + \exp\left\{\left[(3.8 - \gamma_s)^2\right]^{0.16}(3.6 + 0.4c)\right\} \quad \text{kJ/C-mol},$$

where c and γ_s are the carbon number and degree of reduction per C-mol of the substrate (valid for $1 \le c \le 6$). The catabolic reaction (glucose to CO_2) is run just enough to supply D. For glucose and our **K. phaffii** biomass formula (Carnicer et al. 2009) this gives a maximum biomass yield of 0.57 g/g; the measured value is 0.50–0.56 g/g (`data/evidence/hosts.yaml`, grade A). Biomass is counted as ash-free organic matter; ignoring the 7% ash overstates the yield per g of dry cells, but the same convention understates maintenance per g, and the two nearly cancel in glucose per g of cells (audit R1-F3).

Maintenance. In the growth phase maintenance follows Tijhuis et al. (1993), $m_G = 4.5 \exp[-69000/R(1/T - 1/298)]$ kJ per C-mol biomass per hour: 0.0174 g glucose per g cells per hour at 30 °C, close to the 0.0147–0.0161 measured for protein-producing **K. phaffii** in glucose chemostats (Maurer et al. 2006; Rebnegger et al. 2016). In the production phase, where cells barely grow, maintenance falls: Rebnegger et al. (2016) found it threefold lower at near-zero growth. We use the value measured for a protein-secreting strain at low growth, 0.0044 g/g/h (Rebnegger et al. 2024; range 0.0023–0.0161).

Product. The protein's elemental formula comes from its mature UniProt sequence:

- BLG (P02754, 162 residues): $C_{817}H_{1312}N_{206}O_{248}S_9$, 18.3 kDa;
- OVA (P01012, 385 residues): $C_{1901}H_{2997}N_{499}O_{575}S_{22}$, 42.75 kDa;
- bovine lactoferrin (P24627, 689 residues): $C_{3358}H_{5251}N_{945}O_{1002}S_{38}$, 76.1 kDa.

Glycans and phosphates are left out of the formulas: 3–6% of the mass of glycosylated OVA and 5–10% of lactoferrin, which changes the mass balance by less than 1%.

The product reaction makes protein from glucose and ammonia and respires enough extra glucose to pay for synthesis, so the product-reaction yield Y_{ps} fixes its O_2 demand. **No measured BLG or OVA yield on sugar exists**; the design values are lumped yields that already include maintenance and some growth (Intelligen 0.28 g/g; Behm et al. 2022 0.223 g/g). We therefore derive Y_{ps} so that the model reproduces Intelligen's expression phase (their Eq. 2: 100 g glucose ⟶ 28 g OVA + 8 g cells + 4 g host proteins):

$$\frac{1}{Y_{ps}} = \frac{100}{28} - \frac{8/28}{Y_x} - \frac{m_{s,\mathrm{prod}}}{q_p} = 3.571 - 0.501 - 0.919$$

with our growth yield $Y_x = 0.5705$ g/g and Intelligen's specific productivity $q_p = 0.00475$ g/g/h, giving $Y_{ps} = 0.465$ g/g. The growth rate in production (0.0014/h) and the cell density at its start (107 g/L) follow from the same design and its volume balance. With chemostat-level maintenance Intelligen's Eq. 2 would be impossible: maintenance alone would use all of its glucose. The model's production phase then runs at an effective yield of 0.276 g/g, close to Intelligen's 0.28, and the v1 reading (0.28 as the reaction yield, plus maintenance and growth on top) is the low end of the yield range.

Energetic ceiling. Counting ATP gives an upper bound on Y_{ps}: 4 ATP per peptide bond (Martin et al. 2025), paid by respiring glucose at 4 + 12 × P/O ATP per glucose, with P/O = 1.48 for **K. phaffii** (a genome-scale-model value, Chung et al. 2010), i.e. 21.8 ATP per glucose. The ceiling is 0.58 g/g for BLG and 0.59 for OVA. It omits the ATP cost of amino-acid synthesis beyond carbon and electrons (about 5–7% lower by a proxy estimate) and sulfate reduction, so it is an upper bound. The base yield is 80% of it.

Heat. Heat release is computed from formation enthalpies of the dissolved species (about 480–510 kJ per mol O_2). The fermenter's heat check budgets 460 kJ per mol O_2 (Roels 1983, via Almeida Benalcázar et al. 2025; calorimetry gives 455 ± 15, Maskow et al. 2010).

> **Check (reader):** The dissipation formula and its units; the conversion of m_G to 0.0174 g/g/h; the Y_{ps} derivation (`tests/test_model.py::test_y_ps_reproduces_intelligen_eq2`); the ATP ceiling (`src/pftea/product.py`, `energetic_ceiling`). The cost structure rests on these.

2.3 The fed-batch fermenter and its feasibility ceiling

Each production batch has two phases:

1. **Growth** at the maximum specific rate, slowed so that the oxygen uptake rate (OUR) never exceeds what the vessel can transfer and the heat never exceeds what it can remove, up to the production cell density X_{prod}.
2. **Production** for t_{prod} hours at a small growth rate and a constant specific productivity q_p. No product is made during growth, a simplification for constitutive promoters.

The initial liquid volume is solved so that the broth ends at 80% of the vessel, and q_p so that the harvest titer equals the target. A liquid mass balance tracks feed (glucose as 50 wt% solution, ammonia as 30%), O_2 uptake, CO_2 release and evaporation.

Oxygen transfer. The transfer coefficient follows van 't Riet's correlation for non-coalescing broths, $k_La = 0.002(P/V)^{0.7}U_s^{0.2}1.025^{T-20}$ (s^{-1}, SI units), as in Humbird's microbial fermenters, with a log-mean driving force between the bottom (hydrostatic pressure, fresh air) and the top (back-pressure, depleted gas). For each demanded OUR the model picks the air rate and agitation power that minimise compressor plus refrigeration-weighted agitator power, within 5,000 W/m³ and a superficial gas velocity of 0.1 m/s (grade D). In the base vessel, aeration physics alone would allow about 376 mol O_2/m³/h, and the 25 kW/m³ heat-removal ceiling about 186. The lowest of these O_2 limits is the design cap Humbird uses, 150 mol/m³/h (he writes that "200 mol O_2/m³-h is about as high as it gets"); growth runs at it. We apply it per m³ of broth, the usual industrial

basis. Humbird's own workbook applies it per m³ of cell-free liquid, about 97 per m³ of broth at the base cell density; that basis is shown as a sensitivity (Section 4.4).

Other limits. A design is **infeasible**, and never costed as if it worked, when it needs more O_2 or cooling than above, starts below 30% of the vessel volume (the lowest start in a published design, Intelligen's 30.5%), secretes more than 40 mg of protein per g of cells per hour (a judgment, grade D; the base runs at 4.7), exceeds the product's solubility, or has a yield above the energetic ceiling.

Why a ceiling exists. Each kg of BLG made costs about 130 mol O_2 and 9.2 kg of glucose across the batch, including growth and maintenance. The broth must end at 80% of the vessel, so every extra kg of product means more feed and a batch that starts emptier. In the base design that, not O_2, is what caps the titer: at 74 g/L (72 h) the batch would have to start below 30% of the vessel. Longer production phases lower the ceiling (81 g/L at 36 h, 65 g/L at 120 h), because maintenance keeps consuming feed. The limits behind the ceiling are examined in Section 4.4.

2.4 Downstream processing

The train (Figure 1) follows Intelligen's OVA design, with evidence-graded performance (`data/evidence/dsp.yaml`):

- **Cell removal.** Dilute to 45 g/L cells, then two disk-stack centrifuge stages with re-dilution of the paste. Each paste holds 50% extracellular liquid by volume (Petrides: 40–60%), so a fraction $\varphi = eV_p/(V_1 - V_{wc})$ of the product is trapped per stage and the recovery is $1 - \varphi^2$ (96.9% in the base). The wash water is counted.
- **Microfiltration polish** at 20 L/m²/h (grade D, range 10–50) and 97.5% recovery (Behm et al. 2022).
- **Ultrafiltration and diafiltration.** Concentrate at 30 L/m²/h to 50 g/L, diafilter 3 volumes there at 5 L/m²/h (the only diafiltration flux on file, EVERY patent US 2025/0134130), then concentrate to the spray-dryer feed at the same low flux. Rejection 0.995.
- **Dryer feed.** Dairy plants feed high-protein whey concentrates to the dryer at 27–37% solids by membranes, not evaporators (Tetra Pak); egg white is ultrafiltered to about 23% (Tsai et al. 1977). We use 30% for BLG and 23% for OVA.
- **Spray dryer** at 4.87 MJ per kg water evaporated (industry survey, grade A); purchase cost from a supplier-based SuperPro model (Yver et al. 2012).
- **Isolate only:** ammonium-sulfate precipitation (350 g/L; 89% recovery), washed basket centrifuges; the dryer is fed the 20%-solids cake.
- **Lactoferrin only:** cation-exchange capture between microfiltration and ultrafiltration, as in the dairy process and every commercial PF lactoferrin (Section 4.9).
- **Cleaning in place** (7.3 kg solution per kg product, Intelligen) and **spent-cell disposal** ($343/t dry cells in the US, a landfill tipping fee; $158/t in the EU, an anaerobic-digestion gate fee).

The target protein share of the concentrate powder is 85% on a dry basis (grade D; the GRN 1056 minimum is 56%). The base overall recovery is 90%.

OVA solubility. Intelligen stops its ultrafiltration at 50 g/L, "its practical solubility limit", without a source; vendor data sheets give 40–50 mg/mL as quality-control concentrations. OVA dissolves at 200 g/L in water (Wang et al. 2025) and is processed at about 110 g/L in ultrafiltered egg white (Tsai et al. 1977). We therefore do not treat 50 g/L as a limit, and show it as a sensitivity. No data exist at pH 5, the fermentation pH, which is close to OVA's isoelectric point.

2.5 Capital: two conventions

Price-anchored capital (Humbird method; base).

- Purchased equipment costs come from Humbird's cost curves (Aspen ACCE quotes, 2018 dollars); above his largest fermenter quote (196 m³) the curve continues with his own power-fit exponent, 0.69.
- Costs are escalated to 2025 with a factor of 1.33 (CEPCI and the BLS machinery price index; grade B, range 1.23–1.52).
- Installation, buildings, engineering, fees and contingency follow Humbird's "commodity aerobic" facility class, a food-grade plant with the process in a sanitary building. The effective Lang factor (investment over equipment cost) is 5.2.

- Maintenance is 4% and insurance plus local tax 5% of total capital investment (TCI) per year.
- Basis and weakness: with these factors Humbird's baker's-yeast workbook reproduces a market price (anchor 1). That is one product, and the margin behind the "market-implied cost" is his assumption.

Design-report capital (NREL-style; sensitivity). The convention of NREL's 2011 design report, which most published PF studies resemble:

- vendor-quote fermenters: NREL's 300 m³ vessel with agitator costs $0.99M (2007$), 2.9× less than Humbird's curve;
- NREL's installation factors (1.7 for fermenters, 1.8 otherwise) plus indirect costs of 60% of direct cost; no separate warehouse;
- maintenance of 3% of installed process-area cost and insurance of 0.7% of fixed capital.

A Netherlands plant costs 1.09× the same plant in the US Midwest (IEAGHG 2018, same plants costed in both places), applied to capital in both conventions.

2.6 Operating costs, labour and money

Operating costs. Raw materials, utilities and consumables come from the mass and energy balance and the prices in Appendix A: EIA, Eurostat and BLS statistics, trade unit values and published designs (grades A–B).

Labour. Humbird's operator-attention model: operators per fermenter by the fraction of time a batch needs attention, fixed staffing for continuous areas, 4.5 shifts per position, one supervisor per ten staff at 1.4× salary, 100% overhead. The base plant employs 122 people.

Headline metric. MSP_{CCF}: annual operating cost plus a capital charge, divided by annual output. The charge is the capital recovery factor $i/(1-(1+i)^{-n}) = 0.146$ per year at $i = 7.5\%$ and $n = 10$ years (Humbird). It is a cost with a return on capital, not a market price. We also compute the cost of production (COP: operating cost plus depreciation, no return) and an NREL-style discounted-cash-flow price MSP_{DCF} (10% IRR, 30 years, 21% tax, 7-year MACRS, construction over three years, working capital). For base BLG these are $25.5 and $28.5.

2.7 Evidence grades, scenarios and uncertainty

Every parameter and evidence item is graded (Table 3). The four scenarios combine grades differently and are never averaged:

- **academic:** the best public strain performance, read in full text. BLG: 13.88 g/L in **K. phaffii** (Yao et al. 2026), whose production phase runs on ethanol with glutamate feeding; the model has one carbon source and prices it as glucose (per electron the two cost about the same), so this case is flagged. On glucose, Li et al. (2026) reach 5.14 g/L. OVA: 2 g/L in **T. reesei** (Aro et al. 2023; an upper bound for intact OVA, since it includes a clipped form), with no growth in production. Yield and maintenance keep the base values: no measured product yield exists.
- **industry:** 42 g/L, the average that Hawkwood computed over private companies' models of unspecified secreted proteins for GFI (2025) (grade C), with everything else at the base case.
- **required:** the inverse problem: the titer needed for $10, $20 and $50/kg at each production time.
- **physical limit:** product yield at 0.55 g/g (about 95% of the energetic ceiling), the lowest measured maintenance in both phases, no growth in production, 400 m³ vessels with O_2 transfer capped at 250 mol/m³/h (head pressure; grade D) and heat removal at 40 kW/m³, contract sugar and 100 kt/yr. A search over cell density and production time puts the titer at its feasibility ceiling, where cost is lowest, and reports which limits bind.

Grade	Meaning
A	Measured, primary source or official statistic (peer-reviewed data, EIA, Eurostat, BLS, USDA)
B	Design value or patent example from a named source (NREL, Intelligen, Humbird, patents)
C	Company claim, press release, industry survey, secondary transcription
D	Analogy, judgment or calculation from graded inputs
E	Guess; listed in Section 6

Table 3: Evidence grades (`data/evidence/README.md`).

Uncertainty. Two tools:

- **Tornado.** Each parameter with a graded range moves to its low and high value, one at a time. An end that breaks a physical limit is moved to the last feasible value, so no bar quotes the cost of an impossible plant. The capital convention is shown as its own bar.
- **Monte Carlo.** Triangular draws over each scenario's uncertain parameters, with host-specific ranges. Draws that break a physical limit, or for which no plant can be built, count as infeasible; percentiles describe feasible plants only, and the feasible fraction is reported. Draws are independent: no data exist to set correlations, and the feasibility filter is the only coupling.

3 Calibration

The structure was frozen first. Each anchor then changes only its own process inputs, prices, cost year and financing, and misses are recorded in `docs/VALIDATION.md`. `tests/test_calibration.py` pins each recorded value to within five points, so a change that moves an anchor cannot pass silently.

Anchor (type)	Quantity	Ours	Theirs	Diff.
1. Humbird baker's yeast (software + reality)	cost, $/kg dry cells (2018)	1.890	1.890	+0.00%
2. NREL 2011 on-site cellulase (consistency)	glucose, kg/kg protein	3.87	4.17	−7.1%
	O_2, mol/kg protein	83.1	84	−1.0%
	electricity, kWh/kg	5.4	9	−40%
	cost, $/kg protein (2007), price-anchored	10.78	4.24	+154%
	cost, design-report (NREL's own) capital	5.75	4.24	+36%
3. Intelligen 2025 OVA (consistency)	cost, $/kg powder, price-anchored	49.47	43.52	+14%
	total capital investment	$147M	$92M	+60%
	glucose, kg/kg powder	10.5	13.0	−19%
	electricity, kWh/kg powder	10.1	27.1	−63%
4. Voutilainen 2021 recombinant case (consistency)	sugar, kg/kg protein (by construction)	6.37	6.15	+3.7%
	cost, €/t protein, their DCF, price-anchored	21,879	9,007	+143%
	their fixed costs, price-anchored capital	15,612	9,007	+73%
	their fixed costs, design-report capital	11,097	9,007	+23%

Table 4: Calibration anchors. Anchor 1 is Humbird's own workbook, ported line by line; anchors 2–4 are other models run on their own inputs. Diagnoses: `docs/VALIDATION.md`.

What the anchors say.

1. **The port is exact.** It reproduces Humbird's baker's-yeast cost to 0.004%, and that cost sits on the market-implied $1.90/kg (with the caveat that the 33% margin is Humbird's assumption).
2. **Sugar and O_2 agree with the design that states them independently** (NREL, within 7% and 1%). Voutilainen's sugar is matched by construction, and Intelligen's expression phase is, too, because our product yield is derived from it; its growth phase uses 0.40 g/g, below our 0.53 effective and the measured 0.50–0.56. **Electricity does not agree**: ours is 40–63% lower, because our aeration is power-optimised while NREL sizes agitators for peak load and Intelligen aerates at 1.2–1.6 vvm and runs a fertiliser section. These are consistency checks, not calibration fits.
3. **Total costs differ by convention, not biology.** On the same process the published costs are 2.4–2.5× lower than ours (NREL, Voutilainen): vendor-quote fermenters instead of Aspen-based curves; Lang factors of about 3–4 instead of 5; fixed costs of 1.5–2% of capital per year instead of 9%. With their conventions our model still lands above theirs, by +23% (Voutilainen) and +36% (NREL); the rest is mostly labour, buildings and cleaning.
4. **Intelligen's total is within +14%**, with compensating differences: our capital is higher (and our plant needs six whole fermenters where theirs has five; with five the gap is +6.8%), our glucose and electricity lower.

Price-anchored capital is our base because it is the only convention tied to a real product price, and it is the more conservative. Design-report capital is shown everywhere as a named structural sensitivity, not as an error band. NREL itself notes that enzyme producers quoted costs about 47% above its estimate.

4 Results

4.1 Base case

Cost item, $/kg protein	BLG	OVA	Physical intensity, per kg BLG	
capital charge (14.6%/yr)	$12.35	$12.39	glucose	10.3 kg
maintenance	$3.39	$3.40	oxygen	4.7 kg
insurance and local tax	$4.24	$4.25	electricity	9.9 kWh
glucose	$4.10	$4.09	natural gas (drying, steam)	14 MJ
labour	$1.52	$1.52	spent cells (dry)	3.5 kg
spent-cell disposal	$1.20	$1.20	overall yield on glucose	0.098 g/g
electricity	$0.85	$0.86	fermenter volume	660 m³ per kt/yr
media, ammonia, DSP and CIP chemicals	$0.80	$0.81	staff	12.2 per kt/yr
gas, steam, water, membranes	$0.91	$0.93	total capital investment	$848M
MSP_{CCF}	**$29.37**	**$29.45**	production fermenters	33 × 200 m³
per kg powder	$23.72	$23.78	harvest	94 g/L cells

Table 5: Base case: **K. phaffii**, 30 g/L, 72 h production, 10 kt/yr, US Midwest, price-anchored capital.

Capital dominates. Capital charge, maintenance and insurance are 68% of the BLG price; glucose is 14%.

- **Why so much capital.** At 30 g/L the plant needs 33 production fermenters of 200 m³. Each batch occupies a vessel for about 112 h including turnaround and yields 4,299 kg of protein.
- **Why glucose is modest.** The overall yield of protein on glucose is only 0.098 g/g, because the cells, their maintenance and the product reaction all consume sugar; at $0.40/kg glucose it is still only 14% of the price.

The product made per m³ of fermenter per hour is the lever the rest of the paper explores.

4.2 Q1: cost against titer, with every public and claimed data point

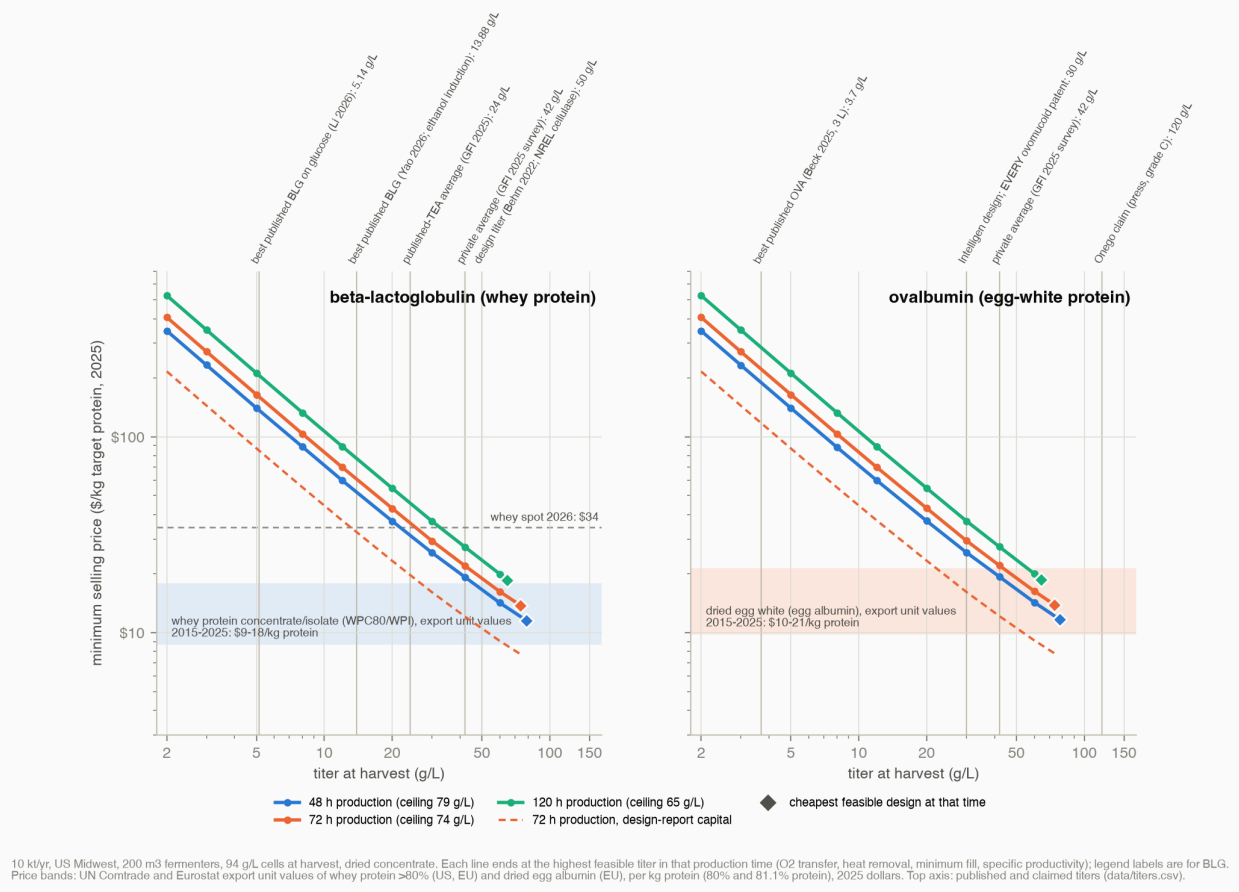


Figure 2: Minimum selling price against harvest titer for three production times. Vertical lines mark the best public titers, the published-TEA and private-sector averages (GFI 2025), and design or claimed titers. Lines end at the feasibility ceiling of the fermenter.

Cost falls roughly as 1/titer until the ceiling (Figure 2):

- **Best public academic strains:** BLG at 13.88 g/L costs $64/kg (ethanol-fed production priced as glucose), and $164/kg at 5.14 g/L on glucose. OVA at 2 g/L in **T. reesei** costs $390/kg. All are several times both incumbents.
- **Private-sector average (42 g/L):** $21.9 (BLG) and $22.0 (OVA), or $12.1 (BLG) and $12.1 (OVA) with design-report capital.

The incumbents. Official trade unit values form a consistent series, deflated to 2025 dollars with the US consumer price index:

- **Whey protein >80%** (WPC80 and WPI mixed; US and EU exports): $9–$18 per kg protein in 2015–2025.
- **Dried egg albumin** (EU exports; 81.1% protein, USDA): $10–$21.
- **2026 whey spike.** WPC80 and WPI rose to about $34 per kg protein at the 2026 peak (USDA Dairy Market News, 2026 dollars). At that price the base case ($29.4) and the industry benchmark ($21.9) would undercut whey. Against 2015–2025 prices, the industry benchmark is above the band with price-anchored capital and inside it with design-report capital.

Parity in production cost is necessary but not sufficient for commercial parity: PF proteins also compete on functionality and price per unit of function.

4.3 Q2: what drives the cost

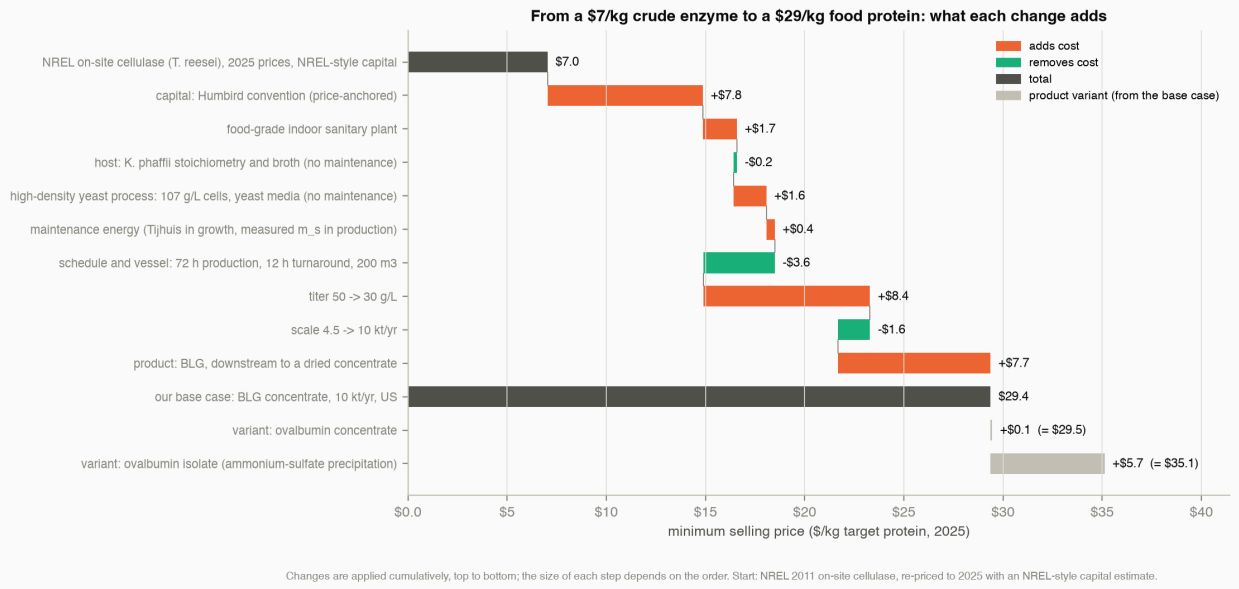


Figure 3: From NREL's on-site cellulase to our base case, one change at a time. The start is NREL's 2011 design re-priced to 2025 with design-report capital.

Why a food protein costs more than an industrial enzyme. Figure 3 walks from an NREL-style crude enzyme ($7.0/kg) to the base BLG concentrate ($29.4/kg). The steps:

- capital: Humbird convention (price-anchored): +$7.8
- food-grade indoor sanitary plant: +$1.7
- host: K. phaffii stoichiometry and broth (no maintenance): −$0.2
- high-density yeast process: 107 g/L cells, yeast media (no maintenance): +$1.6
- maintenance energy (Tijhuis in growth, measured m_s in production): +$0.4
- schedule and vessel: 72 h production, 12 h turnaround, 200 m3: −$3.6
- titer 50 -> 30 g/L: +$8.4
- scale 4.5 -> 10 kt/yr: −$1.6
- product: BLG, downstream to a dried concentrate: +$7.7

The OVA isolate variant adds precipitation and a second recovery loss: +$5.7 over the BLG concentrate. The order of the steps changes their sizes, not the total.

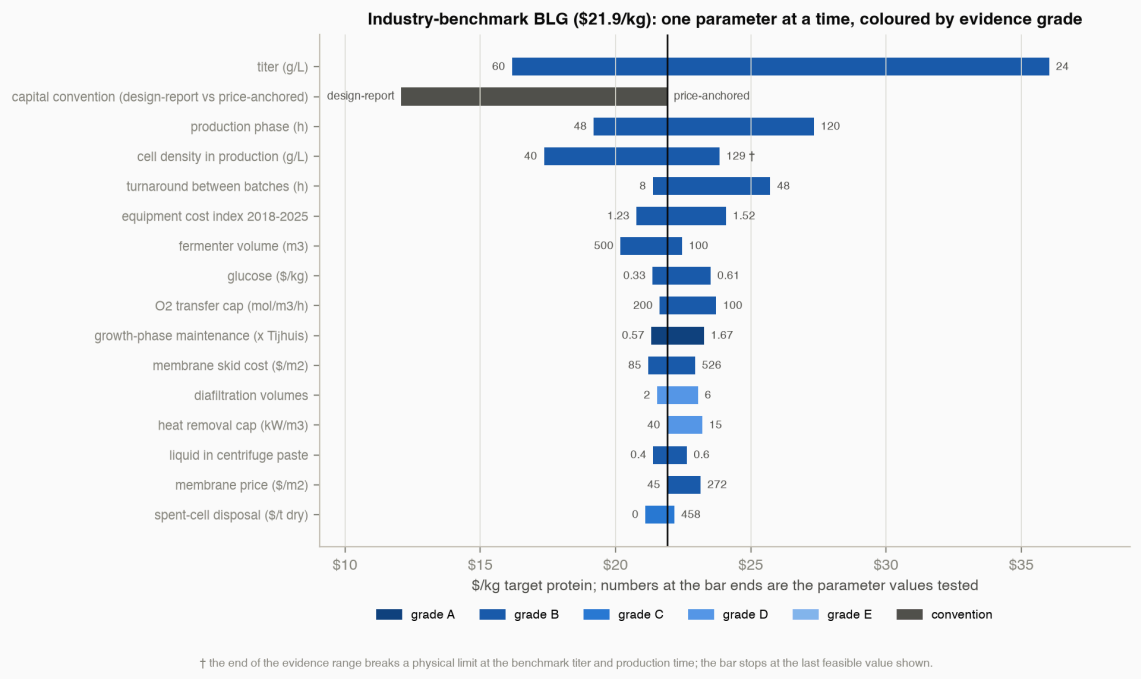


Figure 4: One-at-a-time sensitivity around the industry benchmark (BLG, 42 g/L), coloured by the evidence grade of each parameter's range. † marks an end moved to the feasibility boundary.

Tornado (Figure 4). Around the industry benchmark the largest swings are:

1. titer (g/L): $16.2–$36.0 (grade B)
2. capital convention (design-report vs price-anchored): $12.1–$21.9 (a convention)
3. production phase (h): $19.2–$27.3 (grade B)
4. cell density in production (g/L): $17.4–$23.8 (grade B), one end truncated at the feasibility boundary
5. turnaround between batches (h): $21.4–$25.7 (grade B)
6. equipment cost index 2018-2025: $20.8–$24.1 (grade B)
7. fermenter volume (m3): $20.2–$22.5 (grade B)

What sets fermenter productivity leads: titer, production time and cell density decide how much protein a vessel makes per hour and so how much capital each kg carries. Three points deserve emphasis:

- **Yield matters through the ceiling, not the sugar bill.** At a fixed titer the product yield moves the price by $0.3 (rank 30), because most sugar goes to making and maintaining cells. But it sets the sugar feed (and O_2) each gram of protein needs, and so the ceiling: at 72 h the ceiling is 51 g/L at 0.28, 60 g/L at 0.35, 67 g/L at 0.40, 74 g/L at 0.47, 78 g/L at 0.50, 83 g/L at 0.55 g/g.
- **Vessel size and its cost curve.** 100–500 m³ moves the benchmark by $2.3 (rank 7), and the cost exponent beyond Humbird's largest quote by $0.1.
- **The capital convention** moves it by $9.8, more than any parameter but the titer.

Few strain-performance ranges are grade A; they rest on design values and company-reported averages, because no public BLG or OVA fed-batch dataset reports titer, rate and yield together.

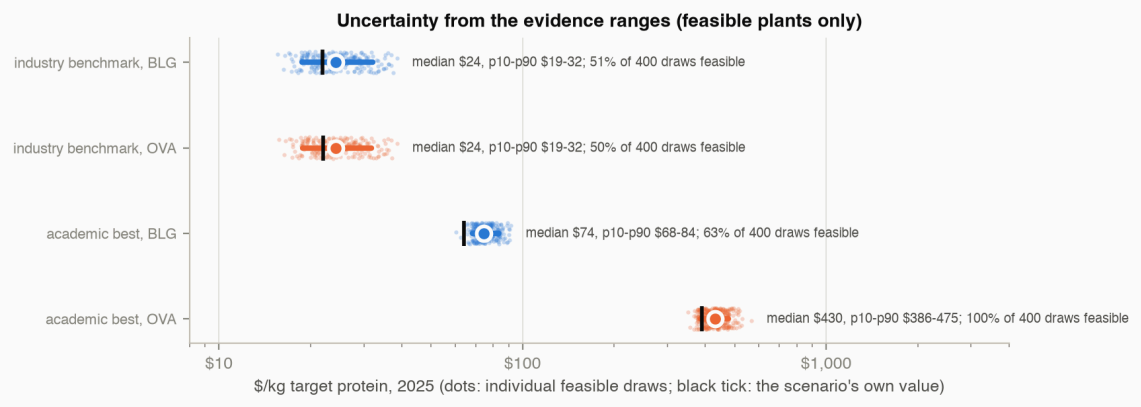


Figure 5: Monte Carlo over each scenario's evidence ranges (triangular), feasible plants only; the black tick is the scenario's own value.

Monte Carlo (Figure 5). Over the evidence ranges the industry benchmark's median is $24 for BLG (p10–p90 $19–$32), and 51% of draws are feasible: combinations of high titer and short production time break the limits. The scenario values sit at these percentiles of their own distributions: industry BLG 35, industry OVA 35, academic BLG 2, academic OVA 12. For the academic strains the p10 is $68 (BLG) and $386 (OVA).

4.4 Q3: what must be true for $10 and $20 per kg

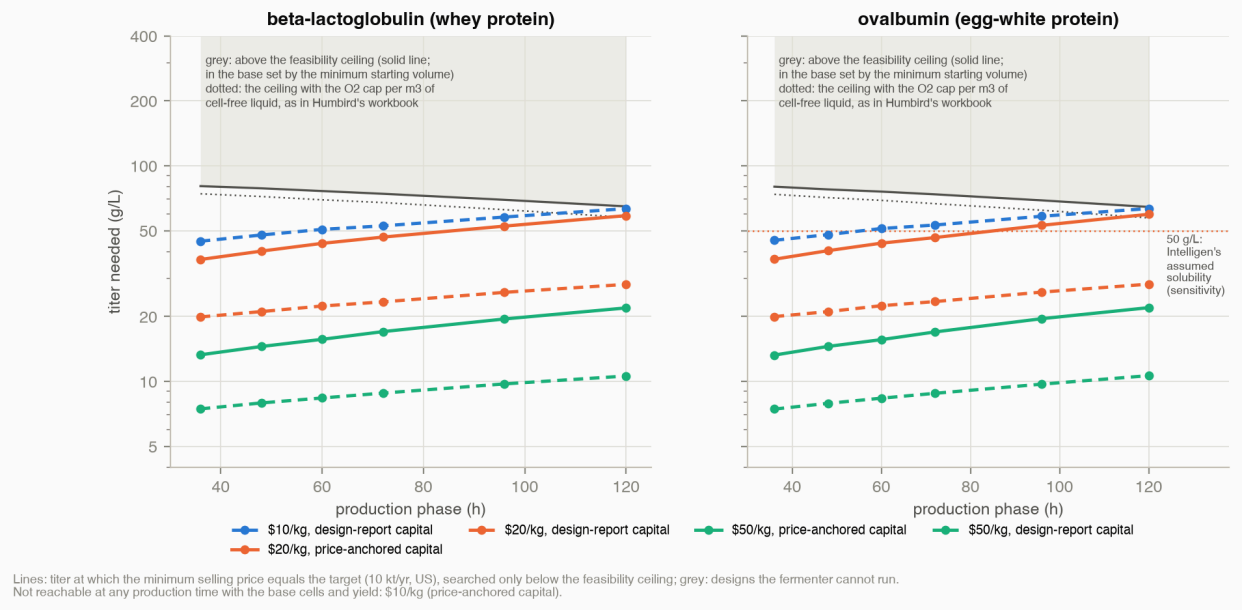


Figure 6: Titer needed for a target price, against production time. Grey: designs the fermenter cannot run. Dotted: the ceiling with the O_2 cap applied per m³ of cell-free liquid, as in Humbird's workbook.

Production phase (h)	36	48	60	72	96	120
Ceiling, BLG	81	79	76	74	70	65
Ceiling, cap per cell-free liquid	74	72	70	68	63	58
$10/kg, price-anchored	—	—	—	—	—	—
$20/kg, price-anchored	37	40	44	47	53	59
$50/kg, price-anchored	13	15	16	17	19	22
$10/kg, design-report	45	48	51	53	58	63
$20/kg, design-report	20	21	22	23	26	28
$50/kg, design-report	7	8	8	9	10	11

Table 6: BLG titer (g/L) needed for each target price, searched only below the feasibility ceiling (first row). — means the target is not reachable at that production time in a 200 m³ vessel with the base cells and yield (`results/analyses.json`, key `required`).

Figure 6 and Table 6 answer the inverse question:

- **$50/kg** needs 13–22 g/L depending on production time: 2.6–4.3× the best public BLG titer on glucose, and well within today's industrial claims.
- **$20/kg** needs about 40–47 g/L in 48–72 h with price-anchored capital, around the private-sector average; with design-report capital, 21–23 g/L.
- **$10/kg** is not reachable with price-anchored capital at the base yield and cell density: the titer it would need lies above the ceiling at every production time. With design-report capital it needs 45–63 g/L in 36–120 h.

How firm is the ceiling? It is a design ceiling, not a law of physics (Table 7). In the base it comes from the vessel's volume: the minimum starting volume (30%, the lowest start in a published design) and the glucose feed concentration (50%, Intelligen's) are both design choices. Relaxing either lifts the 72 h ceiling to about 126 g/L; with no volume limit at all, the secretion ceiling (40 mg/g/h) binds at 129 g/L, and on Humbird's own basis for the O_2 cap (per m³ of cell-free liquid) O_2 transfer binds at 112 g/L. The ceiling also rises with the product yield (Section 4.3) and with lower production-phase maintenance: at 72 h it is 78 g/L at 0.0023, 74 g/L at 0.00437, 68 g/L at 0.008, 55 g/L at 0.0161 g/g/h.

Design choice	36 h	72 h	120 h
base	81 g/L (minimum starting volume)	74 g/L (minimum starting volume)	65 g/L (minimum starting volume)
minimum fill 0.20	90 g/L (secretion ceiling)	125 g/L (minimum starting volume)	118 g/L (minimum starting volume)
glucose feed 65 wt%	101 g/L (secretion ceiling)	126 g/L (minimum starting volume)	119 g/L (minimum starting volume)
no minimum fill	90 g/L (secretion ceiling)	129 g/L (secretion ceiling)	154 g/L (secretion ceiling)
no minimum fill, O2 cap per cell-free liquid	75 g/L (O2 transfer)	112 g/L (O2 transfer)	140 g/L (O2 transfer)

Table 7: BLG titer ceiling (g/L) by production time, and the limit that fails just above it. Base: 30% minimum start, 50 wt% glucose feed, O_2 cap 150 mol/m³/h per m³ of broth.

Titer alone is therefore not enough for $10/kg: yield, cell density, production time, vessel size, the capital convention and scale must move too, which is what the physical-limit scenario does.

4.5 Q4: the physical-limit floor

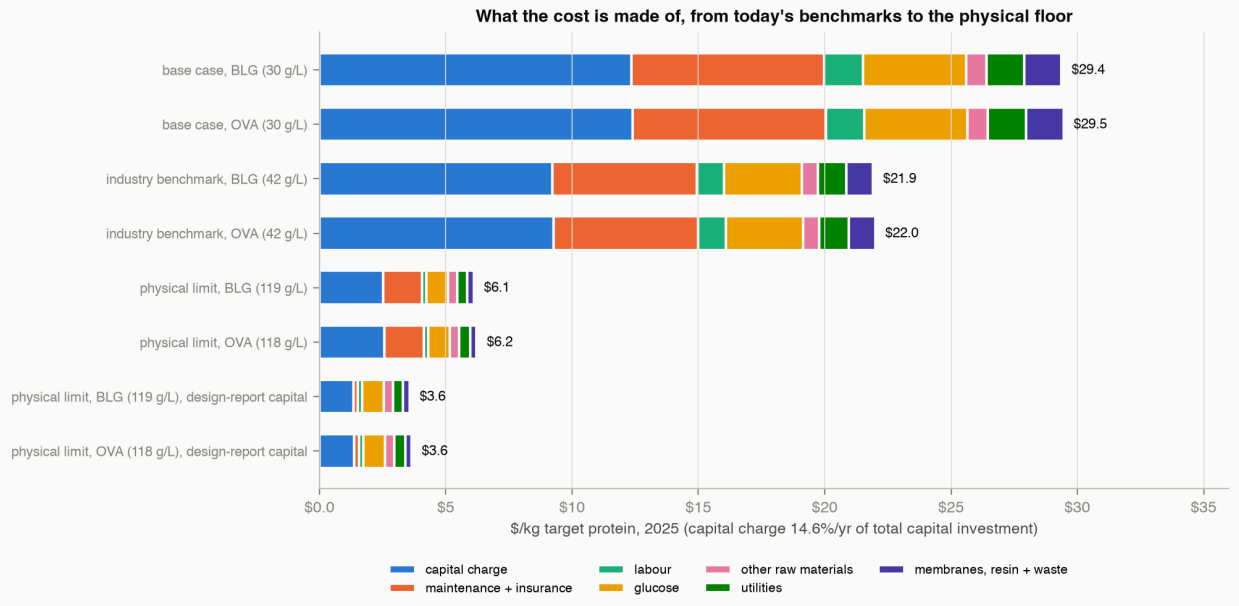


Figure 7: Cost structure from today's benchmarks to the physical limit. The best academic strains are off this scale.

The search (Figure 7) finds the cheapest feasible design. For BLG it has 80 g/L cells, a 72 h production phase and 119 g/L, at a specific productivity of 40 mg/g/h. The limits active there are O2 transfer, specific productivity, minimum initial fill (growth always runs at the O_2 cap; the secretion ceiling and the starting volume stop the titer).

The floor is $6.1/kg BLG with price-anchored capital and $3.6 with design-report capital; for OVA $6.2 and $3.6.

- **Capital still dominates at the floor:** 67% of the BLG floor. Glucose ($0.86) is the largest non-capital item.
- **The secretion ceiling binds** and is a judgment (grade D). At 20 mg/g/h the BLG floor would be $8.2, and at 120 mg/g/h (what our model of NREL's **T. reesei** cellulase implies) $4.6.
- **OVA and the 50 g/L question.** If Intelligen's 50 g/L were a real solubility limit at fermentation pH, the OVA floor would be $10.5 instead of $6.2.

The floor is not a forecast. It is what physics and stoichiometry allow once every biological parameter is at its best measured or theoretical value at the same time. The BLG floor is below the 2015–2025 whey band; the OVA floor is below the egg-white band.

4.6 Q5: company claims against the fermenter's limits

Claim or benchmark	Titer	Modelled as	Shortest feasible production phase	Implied q_p (mg/g/h)	MSP, price-anchored	MSP, design-report
GFI 2025 private-sector average	42 g/L	*K. phaffii*, 107 g/L cells	13 h	38	$14.8	$9.5
GFI 2025 published-TEA average	24 g/L	*K. phaffii*, 107 g/L cells	12 h	21	$23.8	$15.1
Behm 2022 BLG design titer (T. reesei)	50 g/L	*T. reesei*, 35 g/L cells	48 h	40	$13.3	$7.3
T. reesei homologous secreted protein (Behm 2022, grade C)	100 g/L	*T. reesei*, 35 g/L cells	142 h	40	$12.3	$6.3
Onego ovalbumin claim (press, grade C)	120 g/L	*T. reesei*, 35 g/L cells	not reachable	—	n/a	n/a

Table 8: Claimed titers checked against the fermenter's limits in a 200 m³ vessel (O_2 transfer, heat removal, minimum starting volume and the 40 mg/g/h secretion ceiling): the shortest production phase in which each titer is feasible, the specific productivity that implies, and the cost there. The Onego claim concerns OVA at 120 g/L.

Of the 5 claims and benchmarks, 4 are feasible if the production phase is long enough (Table 8). The price of a high titer is time, and time is fermenter capacity, so a claimed titer alone does not settle the cost. The check is physical; whether a strain can sustain the implied productivity is a separate, biological question. Onego ovalbumin claim (press, grade C) cannot be reached at any production phase up to 20 days in our model: either the O_2 demand exceeds what the broth can take up, or the batch would have to start below 30% of the vessel because the feed fills it first. Such claims need better O_2 transfer than our assumptions (for **T. reesei**, a k_La half that of yeast broth, grade D) or a different process (continuous, repeated fed-batch or cell recycle), none of which is modelled here.

4.7 Q6: scale and siting



Figure 8: Industry-benchmark BLG against plant output, US Midwest and north-west Europe, both capital conventions.

Scale (Figure 8). Going from 1 to 10 kt/yr cuts the price by $15.5, and going from 10 to 100 kt/yr saves a further $1.3. Beyond about 10 kt/yr the plant adds 200 m³ fermenters rather than larger ones, so only the downstream train and utilities keep scaling; larger vessels are a separate lever (Section 4.3).

Siting. A Netherlands plant costs $4.1/kg more than one in the US Midwest (base BLG). One at a time, US-level inputs would remove:

- glucose ($0.61 against $0.40/kg): $2.2
- electricity and gas ($0.146 against $0.086/kWh; $19.6 against $4.8/GJ): $0.8
- construction cost (1.09×): $1.8
- wages: nothing (Dutch and Iowa operator pay are within a few per cent).

4.8 Q7: milk and egg equivalents



Figure 9: The cost of replacing the protein of a litre of milk (left) and the egg-white protein of an egg (right) with PF protein, against titer (72 h production, price-anchored capital), beside milk and egg prices in 2025 dollars.

A litre of cow's milk holds 33.7 g of protein, 3.3 g of it BLG; a large egg holds 3.64 g of egg-white protein, 1.96 g of it OVA (`data/evidence/milk_egg.yaml`). At the base cost, the PF protein needed to replace all the protein of a litre of milk costs $0.99, against $0.48 for the litre at the US farm gate in 2025 and $1.07 at retail. Replacing one egg's white costs $0.107, against $0.311 per egg wholesale in 2025, a year of avian-influenza prices ($0.096–$0.311 over 2015–2025).

Dividing a source's price by its protein gives the price per kg of protein at which PF would match it: US farm milk sold its protein at $14–$19/kg in 2015–2025, and wholesale eggs their egg-white protein at $26–$86/kg.

With a 72 h production phase (ceiling 74 g/L), BLG matches the 2025 farm-milk protein price at 70 g/L, and OVA matches the lowest egg-white protein price of the decade at 34 g/L.

These are upper bounds on what the protein is worth. Fat and lactose carry much of the milk price, and yolk part of the egg's. Whey is a by-product of cheese: the whey protein of a litre of milk (6.6 g) is worth $0.057–$0.117 at whey prices, 9%–25% of the farm milk price, so replacing whey protein cuts milk output far less than the "litre of milk" framing suggests. For animal welfare, egg white is the stronger lever: per kg of target protein it involves roughly 50–60 times more animal-years than whey BLG (about 0.6 kg OVA per hen-year against about 35 kg BLG per cow-year). These last two figures are our order-of-magnitude arithmetic, not model outputs.

4.9 Q8: a high-value contrast, lactoferrin

Lactoferrin, an iron-binding milk protein, sells as a supplement and infant-formula ingredient at a hundred times the price of whey. Two recombinant bovine lactoferrins made in **K. phaffii** without methanol have FDA "no questions" letters (GRN 1219 TurtleTree, 2025; GRN 1284 All G, 2026). We model it at 250 t/yr (Intelligen's 2024 design made 251 t/yr; the world market is roughly 0.3–1.6 kt/yr, and sources disagree by five times), with the dairy downstream route: microfiltration, cation exchange, UF/DF and spray drying to 90% lactoferrin. The chromatography step uses a binding capacity of 75 g/L of resin and 16 column volumes of buffer per cycle (EVERY patent), elution at 1 M NaCl, 83.2% step recovery, and resin at $1,250/L (Petrides) lasting 1,000 cycles (industrial lactoferrin columns, Pires et al. 2025). Two guesses (grade E) set the column hardware and the cycle time, which only sizes the resin inventory.

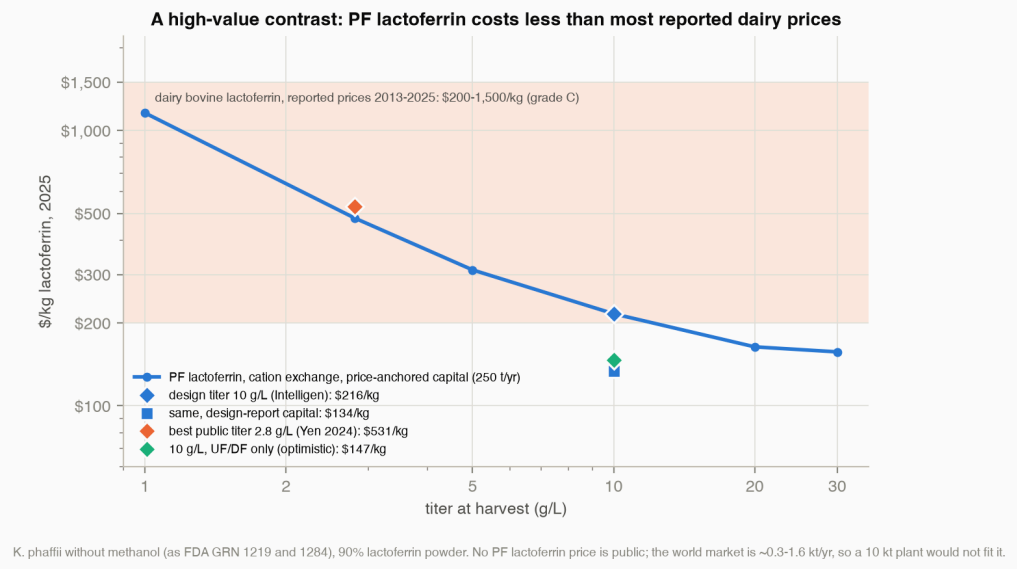


Figure 10: Lactoferrin cost against titer, and the reported dairy lactoferrin price range.

Case (250 t/yr)	Titer (g/L)	Price-anchored	Design-report
Design titer, cation exchange (Intelligen 2024)	10	$216	$134
Best public methanol-free titer (porcine LF, Yen 2024)	2.8	$531	$283
Design titer, UF/DF only (optimistic)	10	$147	$84
Dairy lactoferrin, reported 2013–2025 (grade C)			$200–$1,500 per kg

Table 9: Lactoferrin, $/kg lactoferrin (2025). Resin life is the largest downstream lever: at 125 cycles (biopharma practice) the design case costs $364. Scale: $330 at 100 t/yr, $216 at 250 t/yr, $164 at 1000 t/yr, $158 at 2500 t/yr.

At the design titer PF lactoferrin costs $134–$216 per kg (design-report to price-anchored capital), inside the reported range for dairy lactoferrin, which has a small market and a volatile price (about $200/kg in the 2016 glut, $750–1,500/kg in 2023–2025). At the best public titer it is $531 with price-anchored capital. No PF lactoferrin price has been disclosed, so this is a cost comparison only. The contrast with BLG is the point: the same fermentation and accounting that leave BLG above whey put lactoferrin within or below its incumbent's range, because the incumbent is scarce rather than a by-product.

5 Discussion

5.1 Answers, in brief

- **Q1.** Today's public strains make BLG and OVA at $64 and $390 per kg: not competitive. The private-sector average titer gives $21.9 (BLG) and $22.0 (OVA), or $12.1 and $12.1 with design-report capital. Against 2015–2025 prices, BLG is then above the whey band (inside it with design-report capital) and OVA at the top edge of the egg-white band.
- **Q2.** Fermenter productivity (titer per hour of vessel time) and the capital convention drive the cost; sugar and energy prices come second. The product yield acts mainly through the feasibility ceiling. The strain parameters are the weakest evidence: design values (B) and survey averages (C).
- **Q3.** With price-anchored capital, $20/kg needs 40 g/L at 48 h or 47 g/L at 72 h. $10/kg needs more than titer.
- **Q4.** The physical floor is $6.1 (BLG) and $6.2 (OVA) per kg, bound by O2 transfer, specific productivity, minimum initial fill.
- **Q5.** 4 of 5 claimed titers are feasible in our fed-batch model, some only with long production phases, which cost capital.
- **Q6.** Scale matters up to about 10 kt/yr. EU siting adds about $4.1/kg.
- **Q7.** The base BLG replaces the protein of a litre of milk for $0.99, while the litre costs $0.48 at the farm gate.
- **Q8.** PF lactoferrin already costs less than most reported dairy lactoferrin prices.

5.2 Why published PF costs are lower

Published PF cost estimates for food proteins span roughly $12–39/kg: GFI Europe and Arthur D. Little (2025) give $24–33/kg for their "P1" molecules today and $12/kg for α-lactalbumin after improvements; Intelligen (2025) $39/kg for an OVA isolate at 1 kt/yr. Where we can reproduce the process (NREL, Voutilainen), the gap to our base is mostly accounting: vendor-quote fermenters, lighter installation and indirect factors, and fixed costs of 1.5–2% of capital instead of 9%.

Neither convention is "correct". The price-anchored one reproduces the price of one fermented commodity; the design-report one matches how process-design studies quote greenfield enzyme plants. We read our base as a conservative central estimate and design-report capital as an optimistic one, not as a confidence interval.

5.3 For funders

In order of expected cost reduction per unit of evidence gained (our judgment):

1. **Public fed-batch data.** Titer, productivity and yield measured **together** at 1–10 m^3 for BLG and OVA; no public dataset reports all three, and the product yield is currently derived, not measured.
2. **Productivity and the ceiling.** Near the ceiling, extra titer helps only if it comes without extra time and extra feed; the minimum workable starting volume, the most concentrated feed and O_2 transfer above 150 mol/m^3/h at production scale are the numbers that move it.
3. **Real capital costs.** The capital convention is the largest structural uncertainty; published costs of recent PF plants would settle it.
4. **Downstream data.** Diafiltration flux at scale (one patent value, 5 L/m^2/h, sizes most of the membrane area) and dryer-feed solids for PF proteins.

On the evidence here, PF BLG undercuts whey on production cost in spike years like 2026. At the private-sector average titer with design-report capital ($12.1) it would have undercut whey in 9 of the 21 annual export prices of 2015–2025; with price-anchored capital ($21.9) in 0. High-value proteins such as lactoferrin already cost less than most reported incumbent prices; OVA sits closer to parity than BLG because egg white is dearer than whey.

> **Check (reader):** A judgment, not a model output, written by the AI assistant and not yet endorsed by the author (please revise): the probability that a PF plant built with 2026 technology makes BLG below $10/kg protein is about 0.15, with low confidence. The physical-limit case shows $10 is physically possible, but only if every lever moves at once.

5.4 For EU policy

At Dutch sugar, energy, wage and construction prices the base BLG plant costs $4.1/kg more than in the US Midwest. Cheaper capital is the biggest single lever the model can show: a capital charge at 4% over 20 years instead of 7.5% over 10 cuts $6.7 from the EU cost (more than the whole transatlantic gap), and 4% over 10 years cuts $2.1. US-level glucose would remove $2.2 and US-level energy $0.8. These are cost effects on one plant; the model says nothing about which instruments would deliver them.

5.5 What would change these conclusions

- A peer-reviewed BLG or OVA fed-batch at ≥40 g/L in ≤48 h with a measured yield would test the derived product yield and the ceiling.
- The ceiling rests on two design choices, a 30% minimum starting volume and a 50% glucose feed; industrial practice with lower starts or more concentrated feeds would lift it, and then the secretion rate and O_2 transfer (150–200 mol/m³/h, on either basis) would set it.
- Real capital costs of recent PF plants (for example Liberation Labs' 600 m³ Richmond plant) would settle the convention question.
- Measured OVA solubility at pH 5.

6 Limitations

- **Strain parameters.** Few are grade A. The product yield is derived from a design's lumped stoichiometry, not measured; specific productivities are design values or survey averages. 10 titer rows in `data/titers.csv` are not yet verified in full text.
- **One carbon source.** Methanol-induced and ethanol-fed production phases are priced as glucose.
- **The feasibility limits are design choices**: the minimum starting volume (30%) and feed concentration (50% glucose), which set the base ceiling; the O_2 cap and its basis (150 mol/m³/h per m³ of broth); the heat-removal ceiling (25 kW/m³, grade D); and the secretion ceiling (40 mg/g/h), a judgment.
- **No product during growth**, and constant specific productivity in production.
- **OVA solubility** at fermentation pH is unmeasured; 50 g/L is shown as a sensitivity.
- **Continuous processes** and cell recycle are not modelled.
- **Chromatography** is modelled only for lactoferrin, with guessed column hardware and cycle time.
- **EU wastewater** is priced as in the US; the Dutch levy is per kg of oxygen demand, which is not modelled.
- **Humbird's compressor formula** overstates compressor power by about 1.6×; we keep it for fidelity (about $0.1/kg).
- **Guesses (grade E)** remain for: `hosts.treesei.mu_max_per_h`, `dsp.cex_cycle_h`, `dsp.cex_hardware_usd_per_L_resin`, `utilities.sterilizer_peak_factor`, `utilities.boiler_peak_factor`, `labour.fermenters_per_continuous_crew`, `capital.dsp_area_factor`, `capital.compressor_building_m2_each`.

7 Scoring the pre-registered priors

The priors were written in `docs/SCOPE.md` by the AI assistant (Claude) on 2026-09-26, before any model output existed, and before the author had seen them. They were committed to git together with the first outputs, so the repository cannot prove the order; the session record shows it but is not public.

Prior (2026-09-26)	Stated	Outcome (v2)	Score
1a. Intelligen configuration, $/kg OVA (80% interval)	$38–60	$52.6/kg OVA	inside
1b. Base case, $/kg protein (80% interval)	$28–50	$29.4 (BLG), $29.5 (OVA)	inside, near the low end
1 (superseded). Purified BLG/OVA at ~30 g/L and 1–10 kt/yr	$20–40	10 kt/yr: $29.4 concentrate, $35.1 OVA isolate; 1 kt/yr: $52.6 (Intelligen configuration)	inside at 10 kt/yr, outside at 1 kt/yr
2. P(physical floor < $10/kg)	0.65	BLG $6.1 (yes); OVA $6.2 (yes)	Brier 0.12 / 0.12
3. P(a public BLG/OVA yield on sugar turns up)	0.20	no measured yield; design values exist (Behm 0.223, Intelligen 0.28 lumped)	Brier 0.04 (measured); 0.64 if design values count
4. P(first-try calibration within tolerance): yeast / NREL / Intelligen	0.6 / 0.6 / 0.5	yes / no (+136%) / yes (−3.6%)	Brier 0.16 / 0.36 / 0.25
5. P(public v1 by 2027-04-30 / 2027-07-31)	0.5 / 0.8	open	—

Table 10: Pre-registered priors and outcomes. Prior 4 is scored on the first run (2026-09-26), before the changes recorded in `docs/VALIDATION.md` and `docs/AUDIT.md`. Prior 3 meant a measured strain yield; both readings are scored.

The interval priors were well placed but wide, and the base case landed near their low end. The superseded prior was too optimistic at 1 kt/yr, as its replacement said. The calibration prior was about right on average; the NREL miss came from capital conventions, which it did not anticipate.

8 Reproducibility

```
.venv/bin/python -m pytest                      # model, calibration and invariant tests
.venv/bin/python scripts/run_analyses.py        # all results -> results/analyses.json
.venv/bin/python scripts/make_figures.py        # docs/figures/*.png
typst compile --root . report/pf_tea_report.typ # this document
```

Inputs are in `data/pf/base.yaml` (every value with a range, grade and source), with scenario overrides in `data/pf/scenarios/`. The evidence files are `data/evidence/*.yaml`, with verbatim quotes and locators. Calibration cases are in `data/calibration/`. The audit is `docs/AUDIT.md`.

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A Parameters (generated from `data/pf/base.yaml`)

Parameter	Value	Range	Gr.	Source
plant.product	blg			blg | ova
plant.spec	concentrate			concentrate (GRN 1056-type, UF/DF + dry) | isolate (adds precipitation) | cex (adds cation-exchange capture; lactoferrin) | crude (whole broth)
plant.annual_target_protein_t	10000	1000–100000	C	gfi2025costs Table 2: private-sector volumes 2,500-25,000+ t/yr; SCOPE base
plant.operating_days	330	320–350	B	NREL 2011 uses 8,410 h/yr (96%); Humbird A.1 350 d; Intelligen ~342 batches x 24 h; 330 d allows for campaigns and cleaning
plant.location	us_midwest			us_midwest | eu_nw
strain.host	kphaffii			
strain.titer_g_L	30	5–60	B	intelligen_ovalbumin2025 design basis 30 g/L; Every2024-HY ovomucoid 30.19 g/L (patent Ex. 9); gfi2025costs public avg 24, private avg 42. Tornado ends beyond the feasibility boundary are truncated to it
strain.t_prod_h	72	48–120	B	intelligen_ovalbumin2025 p.6: expression phase 72 h; NREL 2011: 96 h production
strain.X_prod_g_L	107	40–150	B	derived from intelligen_ovalbumin2025: cells at the start of expression = 92 g/L x 107 m3 at harvest (p.7) - 8/28 x 3,210 kg rOVA made (Eq. 2) = 8,927 kg, in 107 m3 less the expression-phase additions (11,464 kg glucose as 500 g/L solution 22.9 m3 + 30% ammonia 3.5 m3 - net O2/CO2 exchange 2.2 m3) = 82.8 m3 -> 108 g/L; feeding then dilutes it to 92 g/L at harvest. Range: K. phaffii glucose fed-batch 93-169 g/L at harvest (Shemesh 2024, A); patents 100-200 (Every2024-HY, B); hosts.yaml
strain.X_max_g_L	150	100–200	A	K. phaffii glucose fed-batch 93-169 g/L (Shemesh 2024); typical 150; patents up to 200 (hosts.yaml). Growth stops here
strain.mu_prod_per_h	0.0014	0–0.02	B	derived from intelligen_ovalbumin2025 Eq. 2: 917 kg cells made on ~8,930 kg in 72 h (8 g per 28 g rOVA at 30 g/L x 107 m3; 92 g/L at harvest) -> ln(9,844/8,927)/72 = 0.00136/h. Range: 0 (non-growing production) to 0.02 (K. phaffii production-phase growth rates, hosts.yaml; D)
strain.Y_ps_g_g	0.465	0.28–0.55	B	derived so that the model reproduces intelligen_ovalbumin2025 Eq. 2 (expression phase: 100 g glucose -> 28 g rOVA + 8 g cells + 4 g HCP, maintenance included). Glucose per g product in expression = 1/Y_ps + (8/28)/Y_x + m_s,prod/q_p = 100/28, with Y_x = 0.5705 g/g (the model's Heijnen growth yield for K. phaffii), m_s,prod = 0.00437 g/g/h (hosts.kphaffii.m_s_prod_g_g_h) and q_p = 3,210 kg / 675,213 kg.h = 0.00475 g/g/h (Intelligen) -> 1/Y_ps = 3.5714 - 0.5008 - 0.9192 = 2.151, Y_ps = 0.465 (audit R2-Q2/R6-C2: 0.28 is a lumped yield, not a reaction yield). low = 0.28 (the lumped value read as a reaction yield, as in v1); high = 0.55 (~0.95 x the P/O-based energetic ceiling 0.58-0.59, which omits amino-acid synthesis cost; D). Checks: Behm 2022 design 0.223 g/g overall (T. reesei BLG); NREL cellulase 0.24 g/g overall
strain.q_p_max_mg_g_h	40	20–120	D	judgment: a physiological ceiling on specific secretion, applied as a feasibility limit. No K. phaffii measurement near it: secreted-protein q_p in K. phaffii fed-batch is typically 0.1-5 mg/g/h (the base design runs at 4.7). high = our model of NREL's T. reesei cellulase design, which implies ~118 mg/g/h (a model output, not an NREL input); low = judgment. Shown as a sensitivity where it binds (physical limit)

Parameter	Value	Range	Gr.	Source
strain.X0_g_L	5	2–15	D	inoculum 5% v/v (Intelligen p.6) times a seed density of ~100 g/L (judgment; Intelligen gives no seed density, and fermenter.seed_final_X_g_L = 60 would give 3 g/L); Humbird A.1 starts at 15 g/L
hosts.kphaffii.biomass_formula	(table in file)		A	Carnicer et al. 2009 Microb Cell Fact 8:65, Table 4: Fab-producing K. phaffii CH1.665N0.134O0.602S0.0039 (evidence hosts.yaml; re-check table cell)
hosts.kphaffii.mu_max_per_h	0.2	0.15–0.25	A	hosts.yaml: mu_max ~0.2 1/h on sugar vs 0.11-0.12 on methanol
hosts.kphaffii.temperature_C	30		B	intelligen_ovalbumin2025 p.6: 30 C
hosts.kphaffii.maint_factor	1	0.57–1.67	A	growth phase: multiplier on Tijhuis 1993 maintenance (0.0174 g glucose/g DCW/h at 30 C, Humbird A.1 method). Producing K. phaffii in glucose chemostats: 0.0147-0.0161 (Rebnegger 2016 citing Rebnegger 2014; Maurer 2006) = 0.85-0.93x. low = 0.0100 (non-producing strain, Rebnegger 2016) / 0.0174; high = 0.029 (5-copy lipase clone, Nieto-Taype 2020) / 0.0174
hosts.kphaffii.m_s_prod_g_g_h	0.00437	0.0023–0.0161	A	production phase (mu ~ 0): Rebnegger et al. 2024 Microb Cell Fact 23:44, VHH-secreting K. phaffii, glucose, 30 C, fitted over mu <= 0.05/h: 'm S (4.37 +/- 0.27 mg g-1 h-1)'. Maintenance falls at near-zero growth (Rebnegger 2016: retentostat 0.0031 vs chemostat 0.0100 for the same strain). low = 0.0023 (1-copy lipase clone, Nieto-Taype 2020); high = 0.0161 (Maurer 2006 chemostat, producing Fab strain). Research evidence file: placeholders item 13
hosts.kphaffii.po_ratio	1.48		B	Chung et al. 2010 Microb Cell Fact 9:50: 'P/O ratio (1.48 mol ATP/mol O)', a genome-scale-model constraint (hosts.yaml kphaffii_po_ratio_chung2010); no measured P/O found
hosts.kphaffii.kla_factor	1		B	yeast broth treated as non-coalescing water-like (Humbird Fig1.3)
hosts.kphaffii.cell_removal	centrifuge			
hosts.treesei.biomass_formula	(table in file)		B	humbird2011 p.38: T. reesei cell mass CH1.645O0.445N0.205S0.005
hosts.treesei.mu_max_per_h	0.1	0.05–0.2	E	no sourced T. reesei mu_max found (hosts.yaml gap entry); fungi grow slower than yeast
hosts.treesei.temperature_C	28		B	humbird2011 Table 19: 28 C
hosts.treesei.maint_factor	1	0.5–2	D	as for yeast (Tijhuis in both phases; no T. reesei production-phase m_s found)
hosts.treesei.m_s_prod_g_g_h	(table in file)			no evidence: the production phase uses Tijhuis x maint_factor
hosts.treesei.po_ratio	1		B	Castillo et al. 2016 Biotechnol Biofuels 9:252: 'the P/O ratio was set to 1', a metabolic-model assumption (hosts.yaml treesei_po_ratio_castillo2016); no measured value
hosts.treesei.kla_factor	0.5	0.3–0.8	D	viscous mycelial broth lowers kLa vs water-like correlations (humbird2011 used a fungal-specific correlation); evidence file
hosts.treesei.cell_removal	filter_press			
energetics.atp_per_peptide_bond	4		A	Martin et al. 2025 BBA Bioenerg 10.1016/j.bbabio.2025.149564: 'The cost assumed for protein synthesis at the ribosome is uncontested, 4 ATP per peptide bond' (hosts.yaml atp_per_peptide_bond)
energetics.substrate_atp_per_glucose	4		A	textbook stoichiometry: 2 ATP in glycolysis + 2 GTP/ATP from succinyl-CoA in the TCA cycle per glucose. Oxidative phosphorylation adds P/O x 12 (24 electrons = 12 O atoms = 6 O2 per glucose), so 4 + 12 x 1.48 = 21.8 ATP per glucose for K. phaffii. Not re-read for this project
fermenter.volume_m3	200	100–500	B	engineering.yaml: industrial aerobic vessels 200-1,000 m3 (Humbird, B), lysine 500 m3 (B), baker's yeast 200 m3 (B); PF designs 136 m3 (Intelligen, B), 150 m3 (Liberation Labs, C), 80-120 m3 culture volume (GFI

Parameter	Value	Range	Gr.	Source
				Europe, C); Humbird STRFERM quotes to 196 m3 (larger sizes extrapolate with capital.fermenter_exponent_above_table)
fermenter.h_to_d	2		B	Humbird Equip Table10 STRFERM 200,000: 196.5 m3, D 5 m
fermenter.fill_max	0.8		B	Humbird Fig1.3!C8; NREL 2011 80%
fermenter.min_initial_fill	0.3	0.3–0.5	B	lowest start fill in a published design: Intelligen OVA starts the growth phase at 41,523 L in a 136 m3 vessel = 30.5% (p.6); NREL 2011 starts at 50% (p.37, high); Impossible US 9,938,327: 40% of the maximum working volume (= 32% of vessel). No design text states a physical minimum (impeller submergence). A batch that must start below this fill is infeasible
fermenter.pv_max_W_m3	5000	2500–5000	B	Humbird 2021 p.5 'an agitation rate up to ~5,000 W/m3' design limit in high-density fermentation (B); low = NREL 300 m3 T. reesei maximum 600 kW on 240 m3 of liquid = 2,500 W/m3 (B, derived)
fermenter.us_max_m_s	0.1	0.05–0.23	D	judgment: no design text states a superficial-velocity limit. Humbird's own 200 m3 aeration model at OUR 150 runs 0.050 m/s at the top (Fig1.3 row 197, B); NREL averages 0.033 m/s (B). high = Humbird 2021 p.5 'sparge rate up to ~2 vvm' in a 200 m3 vessel (8 m liquid height, top at 1.33 bar, 30 C) = 0.23 m/s (derived); Intelligen runs 1.6 vvm (B)
fermenter.otr_cap_mol_m3_h	150	100–200	B	engineering.yaml: Humbird A.1!C93 design cap 150 (B); '200 mol O2/m3-h is about as high as it gets' (Humbird 2021 p.10, B); >120 needed by new processes (Takors, A); 100 industrial experience (C). Applied per m3 of broth (otr_cap_basis)
fermenter.otr_cap_basis	broth			broth (industrial basis, base) | cell_free_liquid (Humbird A.1 applies the 150 cap per m3 of cell-free liquid, ~110 per m3 of broth at 80 g/L; audit R2-1, sensitivity)
fermenter.heat_per_o2_kJ	460	430–500	A	Roels 1983 via Almeida Benalcazar et al. 2025 (engineering.yaml heat_per_o2_roels_design_value); calorimetry 455 +/- 15, range 430-480 (von Stockar 2010, hosts.yaml oxycaloric_equivalent); Cooney/Shuler-Kargi 500 (high). Used only in the OTR_max heat check; the batch heat comes from formation enthalpies (~480-510)
fermenter.heat_removal_max_kW_m3	25	15–40	D	no production-scale ceiling is published. Evidence: 'several million kcal' per hour from 200-300 m3 SCP yeast fermenters = ~8-30 kW/m3 (US 3,986,934, B, derived); internal coils alone ~17 kW/m3 in a 600 m3 column (Almeida Benalcazar 2025, derived with its U 1.4 kW/m2/K), so external exchangers are needed; 42 kW/m3 claimed at pilot scale with tube baffles (US 4,670,397, B). 25 = 1.2 x the OTR-150 load (19.2 metabolic + ~1.2 agitation)
fermenter.turnaround_h	12	8–48	B	intelligen_ovalbumin2025 (5 fermenters, 24 h cycle, 108 h culture -> ~12 h); Humbird A.1 8 h; NREL 48 h
fermenter.feed_glucose_wt	0.5		B	intelligen_ovalbumin2025 p.6: 500 g/L glucose solution (the model tracks volume at 1 kg/L, so 0.5 gives the right feed volume; audit R6-M2)
fermenter.nh3_feed_wt	0.3		B	intelligen_ovalbumin2025 p.6: 'aqueous ammonia (30% w/w), which also serves as nitrogen source'
fermenter.seed_ratio	0.1		B	NREL 2011 p.39: seed vessels 10% of the next; Intelligen 5% inoculum
fermenter.seed_time_h	24	15–36	B	intelligen 15 h per seed step; NREL 2-day seed cycle incl. cleaning
fermenter.seed_final_X_g_L	60		D	seed grown to high density before transfer
media.salts_kg_per_m3	22.3	14.7–38	B	derived from intelligen_ovalbumin2025 Table 3: 'Basal Salts' 816,341 kg/yr over 342 batches x 107 m3 final broth = 22.3 kg/m3 (audit R6-m1). low = Table 1 basal

Parameter	Value	Range	Gr.	Source
				medium 38 g/L in the 41.5 m3 initial volume per 107 m3 final (14.7); high = Table 1 applied to the final broth (38, the v1 value)
media.salts_price_usd_kg	0.2	0.15–0.4	B	intelligen_ovalbumin2025 Table 3 'Basal Salts' $0.200/kg
media.h3po4_kg_per_kg_cells	0.047		B	intelligen Eq.(1): 1.87 H3PO4 per 40 cells
media.h3po4_price_usd_kg	0.85		B	intelligen Table 3: H3PO4 (85%) $0.850/kg
media.antifoam_kg_per_m3	0.2	0.02–1	B	Behm 2022 Table 2: antifoam 0.2 g/L (T. reesei BLG); US 5,260,202 (Delta Biotechnology): 'Typical total usage rates ... 0.02-0.2 g per liter' in sensor-controlled yeast fed-batch (low); high = NREL 2011 corn oil 0.1% v/v, ~1.0 kg/m3 (P-420 13.4 kg/h over ~12.9 m3/h broth, derived)
media.antifoam_price_usd_kg	1.12	0.5–2	C	prices.yaml: Voutilainen 2021 design value EUR 1,000/t x 1.12 (no market price found); range is a guess
media.methanol_kg_per_kg_product	0			methanol-free base; Intelligen uses 0.28 kg/kg (Table 3: 280 t per 1,000 t). Cost only: its O2 and heat are not added to the stoichiometry
media.methanol_price_usd_kg	0.91		B	intelligen_ovalbumin2025 Table 3: $0.910/kg
dsp.dilution_to_g_L_cells	45		B	intelligen p.7: broth diluted to 45 g/L biomass before centrifugation; the first underflow is re-diluted to 45 g/L
dsp.centrifuge_stages	2		B	intelligen p.7
dsp.paste_extracellular_frac	0.5	0.4–0.6	B	Petrides 2015 s.11.3.1 p.11-11: 'the centrifuge paste contains 40 to 60% v/v extracellular liquid' (dsp.yaml centrifuge_paste_liquid_fraction). Wet cells taken as 30% dry matter (Humbird A.1!C67)
dsp.recovery_cell_removal_override	(table in file)			calibration anchors only: fixes the cell-removal recovery instead of the paste model
dsp.centrifuge_kWh_m3	1.35		B	Humbird SI Equip!AB40 'Power demand 1.35 kWh/m3' (dsp.yaml); Westfalia HSB400 ~1.4 (Milledge & Heaven 2011, C)
dsp.mf_flux_LMH	20	10–50	D	no primary-clarification MF flux on file for yeast broth: EVERY US2025/0134130's 10 +/- 2 LMH is a final bioburden filtration (audit R6-M1, low); microalgae MF 20-100 (Slegers, D)
dsp.mf_recovery	0.975		B	Behm 2022 (T. reesei BLG design): 'microfiltration assuming 100% fungus rejection and 97.5% permeate recovery' (dsp.yaml mf_permeate_recovery_fungal_behm)
dsp.uf_flux_LMH	30	20–50	B	intelligen p.8: 30 L/m2/h concentration flux; Petrides 20-50
dsp.uf_rejection	0.995		B	intelligen p.8: rejection 0.995 for rOvalbumin
dsp.diafiltration_conc_g_L	50		B	EVERY US2025/0134130: diafiltration of rOVA at 50 g/L (dsp.yaml); the protein is diafiltered at this concentration, then concentrated further
dsp.df_flux_LMH	5	3–7	B	EVERY US2025/0134130: DF at 50 g/L '5 +/- 2 LMH' (dsp.yaml); also used for the final concentration to the dryer feed (audit R3-F2)
dsp.diafiltration_volumes	3	2–6	D	salt and sugar washout to meet the protein spec (evidence file); EVERY uses 6-8 DV after resolubilising a precipitate
dsp.dryer_feed_solids_blg	0.3	0.27–0.37	C	whey protein practice (Tetra Pak Dairy Processing Handbook): WPC80 UF retentates reach 27-32% DM and are dried at ~36% ('~36% for WPC80', ch.19); WPI and >60%-protein WPC are brought to 35-37% DM by NF/high-concentration membranes, not evaporators (ch.18). value = UF-only midpoint; high = NF. The final concentration step is costed at the DF flux
dsp.dryer_feed_solids_ova	0.23	0.2–0.3	C	egg-white practice applied to rOVA (close analogy): egg white is ultrafiltered industrially from 11.5% to 23% solids (Tsai et al. 1977 J Food Prot 40:449, 24 plant batches, A as data); UF to 20-30% DM before spray drying (EP 2 926 674 B1 citing DD 290128, B). 23%

Parameter	Value	Range	Gr.	Source
				solids is ~200 g/L protein; OVA dissolves at 200 g/L in water (Wang 2025, A), so the Intelligen/vendor '50 g/L practical solubility' is not a limit (audit P-8)
dsp.dryer_feed_solids_lf	0.27	0.2–0.37	D	analogy: no lactoferrin-specific value; whey-protein UF retentates reach 27-32% DM (Tetra Pak, C), so the UF-only low end is used; range as for BLG and egg white
dsp.cex_dbc_g_L	75	50–100	B	EVERY US 2024/0083940 A1 [0700]: 'The ligand density would accommodate protein binding capacity between 50-100 g protein/L resin' (dsp.yaml iex_binding_capacity_cex_every; a design statement, not a measured dynamic capacity for lactoferrin)
dsp.cex_cv_per_cycle	16		B	EVERY US 2024/0083940 A1 Table 4: equilibration 4 + wash 2-3 + elution 3-4 + CIP 3 + regeneration 3 column volumes (midpoints; dsp.yaml iex_column_volumes_per_cycle_every)
dsp.cex_elution_cv	3.5		B	same Table 4: elution 3-4 CV of '25 mM Citrate buffer with 1M NaCl'
dsp.cex_elution_nacl_kg_m3	58.4		B	1 M NaCl (EVERY Table 4) = 58.44 kg/m3; the dairy process elutes with 10-15% NaCl (Mao 2023), higher
dsp.cex_cip_cv	3		B	EVERY Table 4: CIP 3 CV of 1 M NaOH
dsp.cex_cip_naoh_kg_m3	40		B	1 M NaOH (EVERY Table 4) = 40.0 kg/m3
dsp.cex_regen_cv	3		B	EVERY Table 4: regeneration 3 CV of '20% EtOH and 150 mM NaCl', listed per cycle in 'a typical process' (often only for storage; per cycle is conservative)
dsp.cex_regen_etoh_kg_m3	157.8		B	20% v/v ethanol x 789 kg/m3 (EVERY Table 4)
dsp.cex_regen_nacl_kg_m3	8.77		B	150 mM NaCl (EVERY Table 4) = 8.77 kg/m3
dsp.cex_cycle_h	3.5	2–6	E	guess: ~17.5 CV per cycle (16 buffer + ~1.5 load) at ~5 CV/h, a typical packed-bed flow; sets only the resin volume (capital), not the buffer or resin-replacement costs, which scale with protein throughput / capacity
dsp.cex_recovery	0.832	0.6–0.95	B	EVERY US 2025/0134130 Ex. 5: 'Chromatography purification recovered about 83.2% POI' (dsp.yaml iex_cex_step_recovery_every). low: Helaina's whole LF train 'recovery of the purification is >60%' (Lu 2024, C); high: judgment
dsp.cex_resin_usd_L	1250	100–2000	B	Petrides 2015 p.11-33: protein IEX/HIC resins '$500 to $2,000 per liter' (midpoint); low = polymeric resins 'under $100 per liter' (both dsp.yaml). Industrial lactoferrin uses agarose SP Sepharose Big Beads (Pires 2025), a protein resin
dsp.cex_resin_life_cycles	1000	125–1000	A	Pires et al. 2025 J Chromatogr A 1760:466274 (industrial lactoferrin CEX): 'After over a thousand cycles, the column must be replaced' (a lower bound; dsp.yaml iex_resin_lifetime_industrial_lactoferrin). low = Petrides' biopharma 50-200 cycles (midpoint, B)
dsp.cex_hardware_usd_per_L_resin	1000	300–3000	E	guess: columns, skid and automation per litre of packed resin; no public cost found (lactoferrin research pass). Petrides' membrane-system curve (biopharma) is the nearest analogue. Resin first fill is added to TCI unfactored
dsp.cex_diafiltration_volumes	4		D	derived: 1 M NaCl in the eluate is washed out by DF to ~2% of protein at 50 g/L after 4 volumes (58.4 x e^-4 = 1.1 g/L), meeting ash <3.5% (GRN 1284); 3 volumes would leave ~6%
dsp.cex_target_frac_dry	0.9	0.77–0.94	A	lactoferrin share of the powder: GRN 1284 (All G) batches 93.6-95.2% protein x 95.7-99.4% LF of protein (~0.90-0.94); low = GRN 1219 (TurtleTree) batches 80.5-83.5% protein x >=95% (~0.77-0.79)
dsp.nacl_usd_kg	0.184		A	prices.yaml salt_food_grade_eu_import_unit_value: EUR 0.1629/kg (2025) x ECB 1.1300
dsp.naoh_usd_kg	0.585		A	prices.yaml sodium_hydroxide_eu_import_unit_value: EUR 0.5177/kg NaOH (100% basis, 2025) x ECB 1.1300

Parameter	Value	Range	Gr.	Source
dsp.ethanol_usd_kg	0.915		A	prices.yaml ethanol_undenatured_eu_import_unit_value: EUR 0.8099/kg (2025) x ECB 1.1300
dsp.isolate_uf_end_g_L	50		B	intelligen p.8: UF to 50 g/L before precipitation (their design; not a solubility limit, see dryer_feed_solids_ova)
dsp.precip_recovery	0.891		B	intelligen p.8: 90% precipitated x 99% retained by basket centrifuges
dsp.ammonium_sulfate_kg_per_m3	350		B	intelligen p.8: 350 g/L
dsp.ammonium_sulfate_usd_kg	0.18	0.18–0.33	B	intelligen Table 3: $0.180/kg; EU import UV 2025 EUR 0.19 (A); US import UV $0.33 (A, high)
dsp.ammonium_sulfate_recycled_frac	0			Intelligen sells it on as fertiliser; base assumes no recovery credit
dsp.isolate_cake_solids	0.2		B	intelligen p.8-9: 'a soft cake with a moisture content of approximately 80%', fed to the spray dryer (dsp.yaml precipitate_cake_retention_intelligen; audit R3-F4)
dsp.isolate_wash_volumes	6		B	intelligen p.9: 'The protein cake is then washed with 6 volumes of purified water' (dsp.yaml precipitate_wash_volumes_intelligen); applied to the retentate volume
dsp.dryer_MJ_per_kg_water	4.87	3.4–6	A	dsp.yaml: Baker & McKenzie survey average 4.87 GJ/t water evaporated; milk powder 3.4 (3-stage) to 4.9 (1-stage), Ramirez et al.
dsp.dryer_fan_kWh_kg_water	0.05		A	derived from Baker & McKenzie 2005 (dsp.yaml spray_dryer_sec_survey): average 4.87 GJ/t with a fuel-to-electricity ratio of about 27 -> 0.18 GJ/t = 0.05 kWh/kg
dsp.dryer_recovery	0.98		D	cyclone + bag filter losses (evidence file)
dsp.spray_dryer_cost_2018usd	923650	361740–1377300	B	Yver et al. 2012 Int J Mol Sci 13:240 Table 3: spray dryer $897k (2011$, supplier-based SuperPro model) drying ~790 kg/h water, x CEPCI 603.1/585.7 to 2018$. low = BioSTEAM SprayDryer correlation (CEPCI 567, unsourced in its code) at 790 kg/h, x 603.1/567.3. high = Watson Dairy Consulting (C): mid-size milk dryer GBP 5-10M at 1,000 kg/h powder (~1,000 kg/h water) scaled to 790 kg/h at 0.68, x ECB 2025 1.3189 $/GBP, read as an installed package and divided by the model's TCI/PEC factor 4.6 and the 2018->2025 index 1.33 (audit R3-F5)
dsp.spray_dryer_ref_kg_h	790		B	Yver et al. 2012: 10% WPI feed, ~816 kg/h water less ~30 kg/h to the precipitate (derived)
dsp.spray_dryer_exponent	0.62	0.6–0.68	B	Yver et al. 2012 s.3.2.4: halving the feed concentration (~2.1x the water) needed a 'spray dryer (59% cost increase)' -> ln1.59/ln2.1 = 0.62 (derived). low = six-tenths rule; high = Watson Dairy Consulting bands GBP 5-10M (1,000 kg/h) vs 15-30M (5,000 kg/h) -> ln3/ln5 = 0.68 (C)
dsp.spray_dryer_max_kg_h	10000		C	largest single dryers offered: SiccaDania 200-10,000 kg/h powder, Tetra Pak 15,000, GEA 25,000 (Watson Dairy Consulting supplier table); 10,000 kg/h water is conservative
dsp.concentrate_target_frac_dry	0.85	0.7–0.92	D	target protein share of dry powder after UF/DF of a secreted protein; GRN 1056 minimum is 0.56 (0.70 x 0.80)
dsp.isolate_target_frac_dry	0.99		B	intelligen p.9: 'a purity of 99% on a dry basis' (audit R6-m2)
dsp.powder_moisture	0.05		B	intelligen p.9: 5% moisture; GRN 1056 max 7%
dsp.membrane_price_usd_m2	45	45–272	B	dairy spiral-wound elements: CAD 300 per 6.69 m2 module (Mercier-Bouchard 2017 J Dairy Sci Table 4 fn 3) = CAD 44.8/m2 / 1.2966 CAD per USD (ECB 2017) x CPI-U 2017->2025 1.3134 = $45/m2. high = Petrides 2015 'under $200/m2' for industrial membranes x CPI-U 2015->2025 1.3583. Flat-sheet PES quote $171/m2 (2023$; Chen 2023) lies between

Parameter	Value	Range	Gr.	Source
dsp.membrane_life_y	0.5	0.25–2	B	dsp.yaml: Petrides 2015, industrial membranes last >2000 operating hours (lower bound)
dsp.membrane_skid_usd_m2	263	85–526	B	Mercier-Bouchard 2017 J Dairy Sci Table 4 fn 2: 'capital investment costs (excluding membranes) of Can$321.02/m2 for SW membranes' in a 5,185 m2 dairy MF plant (probably installed) / 1.2966 x CEPCI 603.1/567.5. low = SuperPro help: food and water installations '$70-100/m2' of membrane (midpoint; year unknown). high = 2x, the low end of SuperPro's 2-5x multiplier for units built to biopharma specifications. Ceramic MF would be ~$3,900/m2 (same paper), not used by EVERY's process
dsp.membrane_kWh_m3_permeate	5	1–13	D	dsp.yaml: UF ~13 kWh/m3 permeate (A, upper end); typical crossflow UF lower
dsp.cip_solution_kg_per_kg	7.33		B	intelligen_ovalbumin2025 Table 3: CIP-Acid 3,430,444 + CIP-Caustic 3,901,202 kg/yr for 1,000 t/yr powder (dsp.yaml cip_chemicals_price_intelligen); counted as process water and wastewater
dsp.cip_price_usd_kg	0.023		B	intelligen_ovalbumin2025 Table 3: CIP-Acid and CIP-Caustic $0.023/kg
utilities.electricity_usd_kWh	US 0.0862 EU 0.1456	US 0.072 / EU 0.1297–US 0.1014 / EU 0.1631	A	prices.yaml: EIA industrial average 2025 US 8.62 c/kWh (Iowa 7.20 = low, Illinois 10.14 = high). EU = Eurostat nrg_pc_205 NL band IF (70-150 GWh/yr, the size of a 10 kt plant) 2025 mean EUR 0.1289/kWh excl. VAT x ECB 1.1300; EU range = band IF in BE (low, $0.1297) and DE (high, $0.1631), 2025 means
utilities.natural_gas_usd_GJ	US 4.78 EU 19.61	US 3.34 / EU 12.97–US 6.38 / EU 19.61	A	prices.yaml: EIA US industrial 2025 $5.23/Mcf / 1.036 MMBtu/Mcf (low = Henry Hub $3.52/MMBtu, high = Illinois $7.05/Mcf). EU = Eurostat nrg_pc_203 NL band I4 (0.1-1 PJ/yr) 2025 mean (S1 18.89, S2 15.81) EUR 17.35/GJ GCV excl. VAT x ECB 1.1300 (audit R5-7: NL for gas as for electricity); EU range = band I4 in BE (low, $12.97) and NL (high); DE $17.77, DK $16.81
utilities.glucose_usd_kg	US 0.4 EU 0.61	US 0.33 / EU 0.49–US 0.61 / EU 0.674	B	prices.yaml: GFI/Hawkwood 2025 Midwest DE95 $0.40/kg (B); low = Humbird contract $0.26 (2019) x CPI-U 1.2593 = $0.33; high = IMARC US dextrose Jun 2025 $611/t (C). EU (grade C, proxy; audit R5-2): EU white-sugar ex-factory average 2025 EUR 537/t x 1.13 = $0.61/kg (EC agri-food portal, A as data; sugar used as the EU sweetener price reference, not as feedstock: K. phaffii does not use sucrose). EU low = Comext glucose-syrup export UV 2025 EUR 0.4305/kg as traded x 1.13 (a lower bound on the dry basis); EU high = IMARC Germany dextrose Jun 2025 $674/t (C)
utilities.ammonia_usd_kg	0.61	0.38–0.86	B	prices.yaml: US anhydrous NH3 import unit value 2025 $0.611/kg (UN Comtrade, A proxy; EU import UV EUR 0.53 = $0.60); low = Humbird SI $0.30 (2019) x CPI-U 1.2593; high = USDA Iowa retail 2025 $784/short ton. Priced per kg NH3 contained (fed as 30% w/w)
utilities.water_usd_m3	US 1.5 EU 1.62	US 0.33 / EU 1.62–US 3 / EU 1.62	B	prices.yaml: GFI/Hawkwood 2025 Midwest $1.50/m3 (B); low = Humbird well water $0.26 (2019) x CPI-U 1.2593; high = Intelligen $3/t; EU = VEMW 2025 NL large-user average EUR 1.43/m3 (C; no EU range on file)
utilities.wastewater_usd_m3	1	0.4–4	B	dsp.yaml: $0.2-0.5/m3 below 1 g/L BOD (Petrides); Chicago $0.42/kg BOD implies ~$4/m3 at 10 g/L (A). EU priced as US: the NL levy is per kg COD and effluent COD is not modelled (audit R5-8, a limitation)
utilities.steam_gas_GJ_per_t	3.29		D	physics: LP steam ~2,740 kJ/kg from feed water at 25 C (105 kJ/kg), steam tables, at an 80% boiler efficiency (judgment) = 3.29 GJ of gas per t steam, no

Parameter	Value	Range	Gr.	Source
				condensate return (audit R5-6). Steam price = gas price x this
utilities.steam_usd_t	(table in file)			override only (calibration anchors); otherwise derived from the gas price
utilities.biomass_disposal_usd_t_dry	US 343 EU 158	US 0 / EU 0–US 458 / EU 514	C	spent GM biomass is today mostly sterilised and landfilled (Garnica-Garcia 2026 New Biotechnol, abstract). US = EREF 2024 national average MSW landfill tipping fee $62.28/short ton (Waste Advantage, C) at 20% dry solids (Humbird cream yeast, B) = $343/t DM, excl. transport. EU = WRAP 2024/25 UK anaerobic-digestion gate fee median GBP 24/t wet (C) x ECB 2025 1.3189 at 20% DS = $158/t DM (the Kalundborg route: Novonesis/Novo Nordisk biomass to biogas). low = 0: a free outlet as feed (21 CFR 573.587 allows dried K. pastoris in broiler feed) or fertiliser; credits are possible but unquantified. high: US landfill at 15% DS (undewatered paste) $458; EU WRAP range top GBP 78/t = $514
utilities.rt_refrig	0.2629248030187212		B	Humbird RawUtil!BY4 (1.24 hp/ton, Couper)
utilities.cooling_water_dT_C	8		B	Humbird A.1!C158
utilities.building_W_m2	50		B	Humbird Equip Table9: Sanitary 50 W/m2
utilities.sterilizer_peak_factor	2		E	guess: HTST sterilizer sized for twice the average media + feed flow (sizing only; v1 hard-coded value, audit R3-F13)
utilities.boiler_peak_factor	3		E	guess: boiler sized for three times the average steam demand, SIP being intermittent (sizing only; v1 hard-coded value, audit R3-F13)
labour.operator_salary_usd	US 60430 EU 59150	US 58320 / EU 55250–US 79970 / EU 80870	A	prices.yaml: BLS OEWS May 2025 Iowa 51-9011 chemical equipment operators $60,430 (US mean $60,710; Humbird's basis was $50k in 2018); low = Iowa 51-8091 $58,320, high = US 51-8091 chemical plant operators $79,970. EU (grade B, derived): Eurostat SES 2022 NL ISCO OC8 gross earnings EUR 43,822 x NL manufacturing (C) hourly labour cost 2022->2024 (47.3/42.0) x NL industry (B-E) 2024->2025 (50.7/47.8; C not yet published) x ECB 2025 1.1300 = $59,150 (audit R5-9). EU range = the same chain on DE (EUR 40,934) and DK (EUR 59,910) OC8 earnings
labour.burden	1		B	Humbird A.1!C246: 100% labour burden
labour.shifts_per_position	4.5		B	Humbird A.1!C225
labour.employees_per_supervisor	10		B	Humbird A.1!E242
labour.supervisor_factor	1.4		B	Humbird RawUtil!BG24: supervisors paid 1.4x the operator salary
labour.attention_production	0.1		B	Humbird A.1!C231: 10% of online+offline time
labour.attention_seed	0.1		B	Humbird A.1!C230 (large seed)
labour.worker_efficiency	0.8		B	Humbird A.1!C224
labour.continuous_fte_per_shift	(table in file)		D	Humbird A.1 continuous-area FTE rules (0.25-1 per area) extended to the DSP areas; isolate_extra only for the isolate, chromatography only for the cex route
labour.fermenters_per_continuous_crew	12		E	guess: one crew for the continuous areas per ~12 production fermenters (v1 hard-coded value, audit R3-F13); Humbird's rules are per plant of 6-8 fermenters
capital.cost_index_2018_to_2025	1.33	1.229–1.524	B	prices.yaml: CEPCI 2018->2023 x1.323 (B) and 2025 = 1.016 x 2024 (publisher) -> ~x1.33-1.35; BLS PPI machinery & equipment WPU11 x1.327 (A). low = BLS PPI chemical manufacturing PCU325325 x1.229 (A), the index Humbird's Equip sheet uses (290.2 in 2018); high = PPI pumps & compressors WPU1141 x1.524 (A). Equipment-specific PPIs run higher still: heavy-gauge tanks x1.92, heat exchangers x1.64
capital.location_factor	US 1 EU 1.09	US 1 / EU 1.075–US 1 / EU 1.09	B	IEAGHG 2018/04 (Amec Foster Wheeler), Chapter E Tables 5-6: the same plants costed in the Netherlands and the US Midwest at 3Q2016 prices and 1.13 $/EUR: TPC ratio NL/US 1.087-1.090 (coal cases; value, high)

Parameter	Value	Range	Gr.	Source
				and 1.075-1.082 (gas-turbine cases, whose turbine is world-priced; low). Applied to TCI. A PF plant (imported stainless equipment) may sit at the low end
capital.basis	humbird			humbird ('price-anchored capital', base: ACCE fermenter quotes + Petrides/Humbird factors, anchored to baker's-yeast price) | nrel ('design-report capital': vendor-quote fermenters, NREL 2011 installation + Table 27 indirects, 3% ISBL maintenance + 0.7% FCI insurance): structural sensitivity
capital.nrel_basis	(table in file)		B	humbird2011 App. A: F-400 $3,628,673 + A-400 $5,255,007 for 9 units (2007$) = $987,076 per 300 m3 fermenter with agitator, x CEPCI 603.1/525.4 to 2018$ (audit R4-17/R5-5); exponent 0.7 (Area 300 fermentors); install F-400 2.0 and A-400 1.5, cost-weighted 1.70; other equipment 1.8 (Table 25 skidded equipment). Table 27: site development 9% + piping 4.5% + other 4% of ISBL (the 4% warehouse is inside this, so no warehouse building on this basis); prorateable 10% + field 10% + home office 20% + contingency 10% + other 10% of TDC. Table 31: maintenance 3% of ISBL installed, insurance 0.7% of FCI; maintenance_frac_fci is for other design reports (Voutilainen 1.5% of FCI)
capital.fermenter_exponent_above_table	0.69	0.6–0.998	B	Humbird SI Equip!BM74:BO74: power fit of the STRFERM bare-vessel quotes, exponent 0.69, used beyond the largest quote (196 m3). low = six-tenths rule (D); high = the last quoted segment 100 -> 196 m3 (0.998), the v1 extrapolation (audit R3-F3)
capital.facility_class	commodity_aerobic			commodity_aerobic (food-grade, indoor sanitary) | commodity_anaerobic (outdoor, lysine-type)
capital.facility_classes	(table in file)		B	Humbird Equip Table14 (Petrides classes, as tabulated by Humbird): ISBL non-bioreactor DC x PEC, OSBL DC x PEC, E&C x TDC, fees & contingency x TPC, fermentation-area building class
capital.missing_frac	0.1		B	Humbird 4FB sheets: 10% missing-capital estimate on ISBL and OSBL
capital.packing_factor	0.15		B	Humbird 4FB-020!D232: fraction of area occupied by fermenters
capital.building_usd_m2_level_2018	(table in file)		B	Humbird Equip Table9 (ACCE 2016$ escalated to 2018)
capital.shop_area_frac	0.25		B	Humbird SI 4FB-020 row 237: 'Shop | % of ferm area | 0.25'
capital.warehouse_area_frac	0.25		B	Humbird SI 4FB-020 row 238: 'Warehouse | % of ferm area | 0.25'
capital.dsp_area_factor	0.5	0.3–1	E	guess: downstream + drying building area as a fraction of fermentation area; no source found (placeholders item 2). Check: Petrides Table 11.3 puts buildings at 0.45 x equipment purchase cost for enzyme plants (range 0.1-3.0); design.building_to_pec reports the model's ratio
capital.office_m2_per_employee	15		B	Humbird 4FB-020!D235
capital.lab_m2	400	150–600	D	analogy: Humbird 2021 SI 4FB-020 rows 233-234, a 6.8 kt/yr cultured-meat plant has 441 m2 of labs (cell lab 294 + QC lab 147). low = the QC lab alone; high = judgment
capital.compressor_building_m2_each	20		E	guess: floor area per air compressor (v1 hard-coded value, audit R3-F13)
finance.ccf_rate	0.075		B	humbird2021 p.39: i = 7.5%, n = 10 y -> CCF ~15% TCI/y
finance.ccf_life_y	10		B	humbird2021 p.39
finance.maintenance_frac_tci	0.04		B	humbird2021 p.46
finance.insurance_tax_frac_tci	0.05		B	humbird2021 p.46
finance.dcf_irr	0.1		B	humbird2011 Table 33

Parameter	Value	Range	Gr.	Source
finance.dcf_life_y	30		B	humbird2011 Table 33 (construction 8/60/32% and 7-y MACRS are fixed in finance.py with the same source)
finance.tax_rate	0.21		A	US federal corporate rate since 2018 (NREL 2011 used 35%)
finance.working_capital_frac_fci	0.05		B	humbird2011 Table 33

B Evidence base

Evidence file	A	B	C	D	E	Total
data/evidence/hosts.yaml	62	19	3	4	8	96
data/evidence/dsp.yaml	7	72	5	9	0	93
data/evidence/engineering.yaml	38	37	7	1	0	83
data/evidence/prices.yaml	66	23	12	0	0	101
data/evidence/placeholders.yaml	13	30	13	0	1	57
data/evidence/milk_egg.yaml	46	0	0	0	0	46
data/evidence/lactoferrin.yaml	22	4	6	0	0	32

Table 11: Evidence entries by grade. Each entry carries a source, a locator and, where a document was read, a verbatim quote.